



SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING
(Deemed to be University)

Syllabus for
B.S. (Hons) Computer Science
B.S. (Hons with Research) Computer Science
(Specialization in Artificial Intelligence)
(Applicable for the batch 2023-2024 onwards)

Department of Mathematics and Computer Science

Sri Sathya Sai Institute of Higher Learning

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SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING

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DEPARTMENT OF MATHEMATICS & COMPUTER SCIENCE

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B.S. (Hons.) Computer Science / B.S. (Hons. with
Research) Computer Science
(Specialization in Artificial Intelligence)

(Applicable w.e.f the Academic year 2023-2024)

INTRODUCTION

Department of Mathematics and Computer Science is offering an Undergraduate program in the Major area of Computer Science viz: Three Year B.Sc. Computer Science, Four-year B.S Computer Science (Hons) with specialization in Artificial Intelligence (AI) and Four-year B.S Computer Science (Hons) specialization in Artificial Intelligence (AI) with Research.

The department also offers Minor courses in the following areas :

- Mathematics
- Data Science
- Actuarial Science

Students are admitted to this program after completing the 12th class under CBSE (or equivalent) boards with 10+2 in Mathematics from any state of India. In the first four semesters, essential aspects of Basic understanding of Programming are provided through courses such as Problem solving with Computer, Data Structures and Algorithms, Software Lab in C - Part-I and Part II, Operating Systems, Database Management System, Object Oriented Programming Concepts, Computer Organization System and Software Lab in C++. These courses give a great foundation for students in the area of Computer Science. In the last four semesters, they specialize through an AI stream of courses for an Honors degree. At the end of the Four Year program, students have a better understanding of both Computer Science and Artificial Intelligence.

Students after completing the program either join Research or continue for M.Sc. Computer Science or AI and Computing or some professional courses like M.Tech. or take up jobs in various fields including software development and AI research. The B.S in Computer Science Honors syllabus aims to achieve the following objectives.

Program Educational Objectives (PEOs)

1. Graduates from this program will be well-equipped to pursue advanced studies or research in computer science, particularly in AI, at leading institutions. They will build upon the strong mathematical foundations in AI, ML, and Data Engineering acquired during their studies.
2. They will demonstrate excellent analytical and decision-making skills, allowing them to effectively analyse, visualize, and optimize data and algorithms, excelling in roles such as algorithmic engineer, data scientist, and machine learning engineer.
3. With their expertise in designing software requirement documents, programming, debugging, and testing, graduates will quickly adapt to new programming languages and technologies, becoming valuable assets as software engineers or software programmers.
4. They will leverage their skills and professional experience to contribute positively to the growth of both the program and the department, using their knowledge to foster innovation and excellence by establishing startups in their field.
5. Equipped with a deep understanding of technology and a positive attitude towards societal impact, graduates will drive advancements in technology that not only solve technical challenges but also positively influence society.

Program Specific Outcomes (PSOs)

1. **Software Development and Data-Driven Decision Making:** Graduates will be proficient in using software tools and techniques to develop, test, and maintain high-quality software systems. They will leverage data analysis and modelling to make informed decisions in various domains, such as business intelligence, healthcare, and scientific computing.
2. **Artificial Intelligence Proficiency:** Graduates will develop specialized expertise in artificial intelligence, including machine learning and related algorithms. This in-depth knowledge will equip them to design, implement, and evaluate AI-based systems, allowing them to address complex, real-world challenges in both academic research and industry, such as natural language processing, computer vision, and autonomous systems.
3. **Data Science Expertise:** Graduates will acquire specialized skills in data science, including proficiency in mathematics, statistics, and data analysis techniques. They will be capable of leveraging large datasets, extracting meaningful insights, and applying predictive modelling to solve intricate problems in various domains. This expertise prepares them for roles in academia or industry, where data-driven decision-making is essential.

Program Outcomes (POs)

1. **Problem-Solving and Research Skills:** Graduates will be able to effectively analyse and solve real-world problems using efficient algorithms and programming techniques. They will also possess strong research skills, including the ability to conduct independent research, formulate hypotheses, design experiments, and analyse data to derive innovative solutions.
2. **Innovation, Entrepreneurship, and Lifelong Learning:** Graduates will demonstrate their ability to innovate, lead projects, and apply research insights to practical applications. They will be adaptable to emerging technologies and

possess essential life skills, including communication, teamwork, emotional intelligence, and resilience, to thrive in a dynamic global environment.

3. **Ethical Leadership and Sustainability:** Graduates will understand the societal and environmental impact of their professional engineering solutions and strive for sustainable development. They will make ethical decisions, demonstrate professionalism, and align their work with values such as compassion, integrity, and social responsibility. Additionally, they will understand the ethical use of technology and be guided by strong spiritual values.

PO – PEO mapping

	PSO 1	PSO 2	PSO 3	PO 1	PO 2	PO 3
PEO 1	√	√	√	√		
PEO 2	√	√	√	√	√	
PEO 3	√			√	√	
PEO 4		√	√	√	√	
PEO 5	√	√	√		√	√

Scheme of Assessment:

The examination system consists of two components

1. Continuous Internal Evaluation (CIE)
2. End-of-semester examinations (ESE)

Some subjects are evaluated solely through internal assessments, while others include both internal assessments and end-of-semester examinations.

CIE for each subject consists of quizzes, assignments, written tests, or oral tests. These tests are conducted at regular intervals during the semester. These help the student keep abreast with the pace of the course and to reinforce learning. The variety of tests enables both the students and the teachers to evaluate the learning outcomes and performance in various forms of expression. Each component of CIE has its own role and importance in the overall evaluation.

ESE is conducted once at the end of the semester which is comprehensive in its nature covering the entire syllabus of the course.

The evaluation method for theory courses involves a **50:50** ratio between CIE and ESE and practical courses are evaluated through a combination of a series of evaluations and a summative test. For practicals worth 50 marks, the series of evaluations contributes 35 marks, while the summative test contributes 15 marks towards the final evaluation. In the case of practicals worth 100 marks, the series of evaluations carries a weightage of 70 marks, with the summative test accounting for the remaining 30 marks.

Note: For courses with structures such as (3L+1P), (3L+1T), and (2L+2P), there is no separate evaluation for either practical or tutorial formative assessment.

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SCHEME OF INSTRUCTION & EVALUATION

B.S. (Honors) / B.S. (Honors with Research) in Computer Science

(Applicable w.e.f the Academic year 2023-2024)

Course Code	Title of the Course	Course Category	Course Type	Credits	Hours (L-T-P)	Mode of Evaluation	Maximum Marks
SEMESTER -I							
	General English [#]	AEC	T	3	3-0-0	IE	100
	Another Language [#]	AEC	T	3	3-0-0	IE	100
UCSH-101	Problem Solving with Computer	DSC	T	4	4-0-0	IE	100
UCSH-102	Software Lab in C - Part I	DSC	P	3	0-0-6	I	100
UDSC-101	Calculus	IDM	T	4	3-1-0	IE	100
UCSH-103	Software Lab in Python – I	SEC	P	1	0-0-2	I	50
	Awareness ^{\$}	VBC	T	2	2-0-0	I	50
	Multidisciplinary Course ^{##}	MDC	T	3	3-0-0	I	100
	Total			23 Credits	27 Hours		700 Marks
SEMESTER -II							
	General English [#]	AEC	T	3	3-0-0	IE	100
	Another Language [#]	AEC	T	3	3-0-0	IE	100
UCSH-201	Data Structures and Algorithms	DSC	T	4	4-0-0	IE	100
UCSH-202	Software Lab in C - Part II	DSC	P	3	0-0-6	I	100
UDSC-201	Probability and Statistics	IDM	T	4	3-1-0	IE	100

UCSH-203	Software Lab in Python-II	SEC	P	1	0-0-2	I	50
	Awareness ^{\$}	VBC	T	2	2-0-0	I	50
	Multidisciplinary Course ^{##}	MDC	T	3	3-0-0	I	100
	Total			23 Credits	27 Hours		700 Marks
SEMESTER - III							
	General English [#]	AEC	T	3	3-0-0	IE	100
UCSH-301	Computer Organization and Design	DSC	T	4	4-0-0	IE	100
UCSH-302	Discrete Mathematics for Computer Science	DSC	T	4	4-0-0	IE	100
UDSC-301	Statistics for Data Science	IDM	T	4	3-1-0	IE	100
UCSH-303	Software Lab in Data Visualization	SEC	P	2	0-0-4	I	50
	Awareness ^{\$}	VBC	T	2	2-0-0	I	50
	Multidisciplinary Course ^{##}	MDC	T	3	3-0-0	I	100
	Total			22 Credits	24 Hours		600 Marks
SEMESTER -IV							
UCSH-401	Database Management Systems	DSC	T	4	4-0-0	IE	100
UCSH-402	Object Oriented Programming Concepts	DSC	T	4	4-0-0	IE	100
UCSH-403	Software Lab in C++	DSC	P	3	0-0-6	I	100
UDSC-401	Linear Algebra for Data Science	IDM	T	4	3-0-2	IE	100
UCSH-404	Software Lab in SQL	SEC	P	2	0-0-4	I	50
	Awareness ^{\$}	VBC	T	2	2-0-0	I	50
	Total			19 Credits	25 Hours		500 Marks

SEMESTER -V							
UCSH-501	Operating Systems	DSC	T	4	4-0-0	IE	100
UCSH-502	Computer Networks	DSC	T	4	4-0-0	IE	100
UCSH-503	Software Lab in Java	DSC	P	3	0-0-6	I	100
UDSC-501	Optimization for Machine Learning	IDM	T	4	3-0-2	IE	100
UDSC-502	Data Mining and Machine Learning	IDM	T	4	3-0-2	IE	100
UCSH-504	Skill Enhancement Course	SEC	T	3	3-0-0	I	100
	Awareness ^{\$}	VBC	T	2	2-0-0	I	50
	Total			24 Credits	29 Hours		650 Marks
SEMESTER-VI							
UCSH-601	Software Engineering	DSC	T	4	4-0-0	IE	100
UCSH-602	Design and Analysis of Algorithms	DSC	T	4	4-0-0	IE	100
UCSH-603	Software Lab in Web Programming	DSC	P	4	0-0-8	I	100
UCSH-604	Elective-I	DSE	T	4	4-0-0 (3-0-2)	IE	100
UCSH-605	Mini-Project/ Internship	DSC	PW	2	0-0-4	I	50
UDSC-601	Big Data Analytics	IDM	T	4	4-0-0	IE	100
	Awareness ^{\$}	VBC	T	2	2-0-0	I	50
				24 Credits	30 (31) Hours		600 Marks
	Grand Total			135 Credits	162 (163) Hours		3750 Marks

Note: Students who wish to exit after the sixth semester shall be awarded B.Sc. Computer Science degree. Students who complete 8 semesters of study will be awarded a B.S. (Honours) in Computer Science degree. Students with a CGPA of 7.5 or above up to the sixth semester can earn *B.S. (Honours with Research) in Computer Science* by undertaking a research project worth 12 credits in lieu of 3 electives (4 credits each).

SEMESTER - VII							
UCSH-701	Linux System Programming	DSC	4	T	4-0-0	IE	100
UCSH-702	Elective-II	DSE	4	T	4-0-0 (3-0-2)	IE	100
UCSH-703	Elective-III	DSE	4	T	4-0-0 (3-0-2)	IE	100
UCSH-704	Research Methodology	DSC	2	T	2-0-0	I	50
UCSH-803	Research Project/ Project Work Interim Report	DSE	--	PW	0-0-6	I	50
UDSC-701	Cloud Computing	IDM	4	T	4-0-0	IE	100
	Awareness\$	VBC	2	T	2-0-0	I	50
			20 Credits		26 (28) Hours		550 Marks
Note : ** All the Minor courses will be completed by the 7th Semester							
SEMESTER - VIII							
UCSH-801	Formal Languages	DSC	4	T	4-0-0	IE	100
UCSH-802	Elective-IV	DSE	4	T	4-0-0 (3-0-2)	IE	100
UCSH-803	Research Project/ Project	DSE	12	PW/RW	0-0-16	I	150
	Awareness\$	VBC	2	T	2-0-0	I	50
			22 Credits		26 (27) Hours		400 Marks
	GRAND TOTAL		177 credits				4700 Marks

The Course code and the title of the Course is listed in Annexure – 1.

List of available Multidisciplinary courses is given in Annexure - 2.

\$ Course code and title of Awareness courses are listed in Annexure -3.

Continuous Internal Evaluation (CIE) & End Semester Examination (ESE)

Categories of Courses		Modes of Evaluation	
Indicator	Legend	Indicator	Legend
AEC	Ability Enhancement Course	IE	CIE and ESE ; ESE single evaluation
DSC	Discipline Specific Core	I	Continuous Internal Evaluation (CIE)
DSE	Discipline Specific Elective	only	
SEC	Skill Enhancement Course	E	End Semester Examination (ESE) only
MDC	Multidisciplinary Course		
IDM	Inter Disciplinary	Note: 'I' does not connote 'Internal Examiner' 'E' does not connote 'External Examiner'	
PW	Project Work		
RW	Research Project		
VBC	Value Based Course		

PS: Please refer to guidelines for 'Modes of Evaluation for various types of papers', and 'Viva-voce nomenclature & scope and constitution of the Viva-voce Boards'.

Notes:

- 1. Specialization:** A student can earn the specialization in the degree title by completing any four electives from the stipulated list of specialization electives.
- 2. Mini Project:** The Mini-Project will be undertaken during the sixth semester by the candidate. This could be based on an internship (taken online/on-campus) with an industry or a field work etc., with a mentoring faculty from the department.

Students will be asked to make a presentation along with a submission of the report of the work done towards the end of the second semester i.e., within one week before the last working day of the semester. This will be evaluated internally by a panel of a minimum two faculty of the department constituted by HoD/Associate HoD. The total marks for the mini-project would be for 50 marks which will be equally distributed between the presentation and the (online) report submitted.

3. Research Project (UCSH-803) / Project Work (UCSH-803)

Research Project (UCSH-803): A Student who earns a CGPA of 7.5 and above in the first 6 semesters may take up a research project. The research project is for 12 Credits and evaluated for 200 Marks. Evaluation of the project is Internal by the project committee appointed by the HoD. The Evaluation method is given below:

- Evaluation will be done in THREE parts.
- First Evaluation is through interim evaluation for 50 marks done towards the end of 7th semester.
- The second evaluation component is through a project work presentation by the student to the project committee nominated by the Head of the Department

(HoD/A.HoD) towards the end of the 8th semester, followed by a Viva Voce. This will be evaluated for 100 marks.

- d. Third Evaluation is through the project report submitted by the student to the guide. The project supervisor/an internal member will evaluate the project report for 50 marks.
- e. Project Committee: A student doing a project must have a project supervisor allotted by the HoD. The project supervisor and another one or two members of the Department nominated by the HoD constitute the Project Committee.

Project Work (PW): A Student who does not earn a CGPA of 7.5 and above in the first 6 semesters may take up project work. The project work is for 12 Credits and evaluated for 200 Marks. Evaluation of the project is the same as Research project (RW).

4. **Discipline Specific Electives (DSEs) and Specialization:** The Department offers several interesting electives which are essential for applying Computer Science skills in various domains of IT. Among these DSEs, any one who completes four AI category DSE courses will earn the specialization degree.

DSEs for Artificial Intelligence Specialization

1. **UCSH-AI:** Artificial Intelligence [4 Credits]
2. **UCSH-DL:** Deep learning [4 Credits]
3. **UCSH-MLO:** Machine Learning Operations [4 Credits]
4. **UCSH-NLP:** Natural Language Processing [4 Credits]

5. **Skill Enhancement Courses (SECs):** The department offers a few SECs to develop hands on skills and awareness which can be handy or supplementary to trendy areas in CS such as Data Science or Cyber Security

List of SECs

1. Software Lab in Python-I
2. Software Lab in Python-II
3. Software Lab in Data Visualization
4. Software Lab in SQL
5. Fundamentals of Actuarial Mathematics

Note:

- The Choice of Electives and Streams of Specialization offered in a semester are subject to the discretion of the Head of the Department
- Specified Syllabus & Schema of Instruction and Evaluation for 3rd and 4th years is subject to revision based on future BoS recommendations

Scheme of Evaluation and Instructions for Skill Enhancement Courses

Paper Code	Title of the Paper	Category	Credits	Periods	Modes of Evaluation	Types of Papers	Maximum Marks
Semester I							
UCSH-103	Software Lab in Python-I	SEC	1	0-0-2	I	P	50
Semester II							
UCSH-203	Software Lab in Python-II	SEC	1	0-0-2	I	P	50
Semester III							
UCSH-303	Software Lab in Data Visualization	SEC	2	0-0-4	I	P	50
Semester IV							
UCSH-404	Software Lab in SQL	SEC	2	0-0-4	I	P	50
Semester V							
UCSH-504	Cyber Security	SEC	3	3-0-0	I	P	100
			9 credits	15 Periods			250 Marks

6. Multi-Disciplinary Courses (MDCs): A student is expected to undertake a MDC elective that is outside his major discipline. The Department of Mathematics and Computer Science (DMACS) offers the following MDCs for undergraduate students.

Scheme of Evaluation of Multidisciplinary Course

Paper Code	Title of the Paper	Course Category	Credits	Periods L-T-P	Mode of Evaluation	Max. Marks
	Fundamentals of Statistics with Excel	MDC	3	2-0-1	I	100

	The Language of Mathematics	MDC	3	3-0-0	I	100
	Fundamentals of Actuarial Mathematics	MDC	3	3-0-0	I	100

7. Minor Disciplines: Department of Mathematics and Computer Science (DMACS) offers the following Minor Disciplines for Undergraduate students with required eligibility to earn a Minor Degree:

1. Mathematics
2. Computer Science
3. Actuarial Science
4. Data Science

However, the Minor degree or courses will be offered subject to available logistics such as faculty, timetable slots etc.

Minor Certification: To earn a minor certification in a specific discipline, a student has to earn a minimum of 12 credits in 3 years of B.S program or 16 credits in 4 years of B.S Hons program in that particular discipline. It is to be noted that, the minimum credits earned must be from courses which are of difficulty level 300 or above. A student majoring in a particular discipline cannot obtain a minor in the same discipline.

The courses from any of the Minor disciplines will be offered based on the discretion of the HoD of DMACS.

Syllabus for Core Courses

Semester-I

Course Title	Problem Solving with Computer	L	T	P	C	Periods
Course Code	UCSH-101	4	0	0	4	56
Course Category	DSC					
Course Objectives: This course is designed to introduce students to algorithmic thinking. It introduces the fundamental concepts in algorithm design and how to apply these concepts in order to develop algorithms to solve specific problems.						
Course Outcome: Develop the skill set to 1. Think in the algorithmic way to solve simple computational problems 2. Choose the appropriate control structures necessary to solve specific problems 3. Formulate simple algorithms for arithmetic and logical problems 4. Test and execute the programs and correct syntax and logical errors 5. Implement conditional branching, iteration and recursion 6. Decompose a problem into functions and synthesize a complete program using divide and conquer approach 7. Use arrays and a few structures to formulate algorithms						
Prerequisites for the Course: 10+2 School Level Mathematics						
Unit	Topic	Contents				Periods
1	Introduction to Problem Solving and Fundamental Algorithms	Introduction, Problem Solving aspect, Top-down design, Foundation in efficiency and analysis of algorithms, Concept of a Variable, Iterative computational examples				11
2	Factoring Methods and Array techniques	Square root of a number, Greatest Common Divisor, Prime numbers, Prime factors, Raising a number to a large power, Fibonacci numbers, Simple Array operations, Removal of Duplicates, Array partitioning				11
3	Merging, Sorting and Searching	Two-way merge, Sorting by selection, Sorting by exchange, Sorting by insertion, Sorting by diminishing increment, Binary Search				11

4	Text Processing and Pattern Searching	Length adjustment, Left and right justification, Keyword search	12
5	Recursive Algorithms	Recursive Quicksort, Towers of Hanoi, Sample generation, Combination generation, Permutation generation	11
		Total	56

Key Text:

1. R. G. Dromey, How to Solve it by Computer, Pearson (Indian Subcontinent Low price edition), 2007. Chapters: 1 to 6, 8

Suggested Reading:

2. Jean-Paul Tremblay And Richard B. Bunt, Introduction to Computer Science, An Algorithmic Approach, McGraw Hill, Second International Edition, 1981.

Course Title	Software Lab in C - Part I	L	T	P	C	Periods
Course Code	UCSH-102	0	0	6	3	84
Course Category	DSC					
Course Objectives: This lab course introduces the basics of the C language to students. It would enable a student to implement algorithms for simple problems as programs in C thereby improving algorithmic thinking.						
Course Outcome: Develop the skill set to 1. Formulate the algorithms as C programs for simple problems 2. Translate given algorithms to a working and correct program 3. Ability to correct syntax errors as reported by the compiler 4. Ability to identify and correct logical errors encountered at run time 5. Ability to write iterative as well as recursive programs 6. Ability to represent data in arrays and simple structures						
Prerequisites for the Course: 10+2 School Level Mathematics						
Unit	Topic	Contents				Periods
1	If-else, for & while loops	Basic programs for learning syntax, Maximum of n numbers, Printing different patterns of ‘*’ based triangles				12
2	Iterative Programs	Square root of a number, Greatest Common Divisor, Prime numbers, Prime factors, Raising a number to a large power, Fibonacci numbers, Simple Array operations, Removal of Duplicates, Array partitioning.				18
3	Merging, Sorting and Searching	Two-way merge, Sorting by selection, Sorting by exchange, Sorting by insertion, Sorting by diminishing increment, Sorting by partitioning, Binary Search.				18
4	Recursive Programs	Recursive Quicksort, Towers of Hanoi, Sample generation, Combination generation, Permutation generation				18
5	String Manipulation programs	Length of a string, String comparison, Palindrome check, Case changing, Length adjustment, Left and right justification, Keyword search.				18
		Total				84
References: 1. Richard F. Gilberg and Behrouz A. Forouzan, Computer Science - A Structured Programming Approach using C, ISBN-13: 978-0534491321, ISBN-10: 9780534491321, 2005. 2. Dennis Ritchie, The C Programming Language, 2nd edition, Prentice Hall India Learning Private Limited, 1990						

Semester – II

Course Title		Data Structures and Algorithms		L		T	P	Credits	Periods
Course Code		UCSH-201		4		0	0	4	56
Course Category			DSC						
	Course Objectives: This course is designed to introduce the process of designing data structures to solve fairly complex problems. It imparts the following. 1. Basic concepts of data structures and algorithms 2. Concept and use of lists, trees, heaps and graphs 3. Searching techniques 4. Enable students to write algorithms for solving problems with the help of fundamental data structures								
	Course Outcomes: Develop the skill set to 1. Choose the appropriate data structure for solving a specific problem 2. Analyze the algorithms to determine the time and computation complexity 3. Implement different search techniques 4. Solve multiple problems using lists, trees, heaps and graphs 5. Determine the best suited data structure to solve the problems								
	Prerequisites for the Course: UCSH-101								
Unit	Topic		Contents					Periods	
1	Fundamentals and Linear Lists		Abstract data type, Algorithm Analysis using Big-O notation, Stack,, Queue, Linked List, Circularly linked list, Doubly linked list, Applications					10	
2	Trees		Binary Tree, General Tree, Binary Search Tree, AVL Search Tree, Applications					14	
3	Heaps		Basic concepts, Operations, Algorithms, Implementation, Applications					8	
4	Graph		Basic concepts, Representations, Traversals, Algorithms, Applications					14	
5	Searching		Linear Search and variations, Binary Search, Hashed list search, Collision resolution					10	

			Total	56
	Key Text: Richard F. Gilberg and Behrouz A. Forouzan, Data Structures - A Pseudocode approach with C, second edition, Cengage Learning, 2005. Chapters : 3 – 9, 11, 13 References: Mark Allen Weiss, Data Structures and Algorithm Analysis in C, Second Edition, Pearson Education, 1997.			

Course Title	Software Lab in C - Part II	L	T	P	C	Periods
Course Code	UCSH-202	0	0	6	3	84
Course Category	DSC					
Course Objectives: This lab course introduces the basics of data structures using C language. It would enable a student to design & implement data structures in C to solve fairly complex problems thereby improving data structure design skills.						
Course Outcomes: Develop the skill set to 1. Program basic data structures in C. 2. Implement Lists, Trees, Heaps and Graphs appropriately to solve problems 3. Implement different searching techniques						
Prerequisites for the Course: UCSH-102						
Unit	Topic	Contents				Periods
1	Pointers	Simple programs for learning C Pointer Concepts, Pointers and arrays, passing pointers in functions				6
2	Linked Lists	Implementing Singly linked list, doubly linked list, circular linked list with a few applications				18
3	Trees	Implementing General Tree, Binary Tree, Binary Search Tree, AVL Trees with a few applications				24
4	Heaps and Graphs	Implementing Heaps, Adjacency list representation of graphs, breadth first traversal, depth first traversal, basic functions in graph				24
5	Search	Linear search, Binary search, hashed list search				12
		Total				84
References: 1. Computer Science - A Structured Programming Approach using C by Behrouz Forouzan & Richard Gilberg. 2. The C Programming Language by Dennis Ritchie 3. Yashwant Kanetkar, Pointers in C.						

Semester-III

Course Title	Computer Organization and Design	L	T	P	C	Periods
Course Code	UCSH-301	4	0	0	4	56
Course Category	DSC					
Course Objectives: This course introduces students to the fundamental concepts of computing systems. This is to familiarize students to equip them with an understanding of hardware design including logic design, basic structure, and behavior of the essential functional modules of the computer. It also gives them how the system interacts to accomplish the required processing for the needs of the user.						
Course Outcomes: Develop the skill set to: 1. To know the basics of Instruction Set Architecture (ISA) and organization of Computer components. 2. Solve simple boolean algebra problems as related to designing computer logic, through combinational and sequential logic circuits. 3. Understand how computers represent and manipulate data and convert it between different number systems. 4. Understand the memory hierarchies and processor structure and functions						
Prerequisites for the Course: NA						
Unit	Topic	Content				Period
1	Arithmetic and Logic	Basics: Functional Blocks in a Computer System, Number system and Computer Arithmetic, Boolean algebra, Gates and combinational logic, Sequential Logic circuits.				20
2	Central Processing Unit	Addressing Modes, Instruction Set and Instruction Execution Flow, processor structure and function				12
3	Control Unit	Hardware and Micro-program based control Unit Design				10
4	Memory	Memory Architecture,Cache,Internal memory				12
		Total				56 Hrs

Key Text:

1. William Stallings, Computer organization and architecture: designing for performance, Pearson Education India, 2000.

Chapters : 3 (3.1-3.2), 4 (4.1-4.3), 5 (5.1), 7 (7.1-7.6), 9 (9.1-9.5), 10 (10.1-10.2), 11 (11.1), 12 (12.1 to 12.4), 20 (20.1-20.5)

Suggested Reading:

1. Carl Hamacher, Zvonko Vranesic and Safwat Zaky. "Computer Organization", McGraw Hill, 2011.

2. David A. Patterson, John L. Hennessy. "Computer organization and design: the hardware/software interface". Elsevier, 2011.

Course Title	Discrete Mathematics for Computer Science	L	T	P	C	Periods
Course Code	UCSH-302	4	0	0	4	56
Course Category	DSC					
Course Objectives: It emphasizes conceptual understanding, quantitative reasoning, and contemporary applications by maintaining a dynamic balance among theory, applications, modeling, and drill. Students work to develop a deeper understanding of a range of mathematical topics using interactive math tutorials that allow the student to solve math problems interactively. Students will gain some confidence on how to deal with problems which may arrive in computer science in the near future.						
Course Outcomes: Develop the skill set to 1. Write and interpret mathematical notation and mathematical definitions 2. Improve the proof writing skills 3. Understand the basic mathematical structures, counting principles 4. Solve linear recurrence relations 5. Describe and solve some real time problems using concepts of graph theory						
Prerequisites for the Course: NA						
Unit	Topic	Contents				Periods
1	Logic	Propositional Logic (PL), Applications-Logic Circuits, Propositional Equivalences, Normal forms (PCNF, PDNF), Predicates and Quantifiers, Nested Quantifiers, Rules of Inference; Introduction to Proofs-Direct Proofs, Proofs by Contraposition and Contradiction.				12
2	Counting Principles	The Basics of Counting, The Pigeonhole Principle, Permutations and Combinations, Binomial Coefficients and Identities, (combinatorial approach) Generalized Permutations and Combinations, Recurrence Relations, Modeling with Recurrence Relations, Solving Linear Recurrence Relations.				12
3	Relations	Relations and Their Properties, Representing Relations, Closures of Relations, Equivalence Relations, Partial Orderings, Hasse Diagrams.				10

4	Graphs and Trees	Graphs: Graph and sub graphs, Models, Special Types of Graphs, Representing Graphs and Graph Isomorphism, Connectivity, Euler and Hamilton Paths, Shortest-Path - Dijkstra's algorithm, Planar Graphs- Euler's Formula, Graph Coloring Trees: Introduction to Trees, Properties, Tree Traversal, Spanning Trees (BFS, DFS), Minimum Spanning Trees, Prim's and Kruskal's Algorithms.	14
5	Algebraic Systems	Definitions, Examples and general Properties, Algebraic structure, Semigroups, Monoids, Groups, Subgroups, Homomorphism and Isomorphism.	8
		Total	56
Key Text: 1. Kenneth H. Rosen, Discrete mathematics and its applications, Eight Edition, McGraw Hill, 2019 2. Tremblay J.P and Manohar R, Discrete Mathematical Structures with Applications to Computer Science, TMH Pub. Co. Ltd, 2010.			
Suggested Reading: 1. Kolman, Busby, Ross: Discrete Mathematical Structures, Pearson education, 4th Edition. 2. T. Veerarajan: Discrete Mathematics, with Graph Theory and Combinations(2008), Tata McGraw – Hill Publishing Company Ltd, New Delhi 3. Discrete Mathematics, Dr A Singaravelu et al, Meenakshi Agency.			

Semester-IV

Course Title	Database Management Systems	L	T	P	C	Periods
Course Code	UCSH-401	4	0	0	4	56
Course Category	DSC					
Course Objectives: To understand different issues involved in the design and implementation of database systems.						
Course Outcomes: Develop skill set to 1. Draw ER diagrams for a given problem 2. Design relational database with normalization 3. Handle concurrency control and storage requirements of a database						
Prerequisites for the Course: UCSH-202,UCSH-301						
Unit	Topic	Contents				Periods
1	Introduction	Characteristics of database approach, Purpose of Database Systems-Drawbacks of file System, data models, DBMS architecture and data independence, E-R Modeling: Entity Course Categories, entity set, attribute and key, relationships, relation Course Categories, roles and structural constraints, weak entities, enhanced E-R and object modeling, Sub Classes:, Super classes, inheritance, specialization and generalization				14
2	File Organization	Indexed sequential access files, implementation using B++ trees, hashing, hashing functions, collision resolution, extendible hashing, dynamic hashing approach implementation and performance.				14
3	Relational Data Model	Relational model concepts, relational constraints, relational algebra. SQL: SQL queries, programming using SQL EER and ER to relational Mapping: Database design using EER to relational language, The Unified Modeling				14

		language UML-features of Good Relational Design.	
4	Data Normalization	Functional dependencies, Normal form up to 3rd normal form. Concurrency Control: Transaction processing, locking techniques and associated, database recovery, security and authorization. Recovery Techniques, Database Security	14
		Total	56 Hr
Key Text: 1. R. Elmasri and SB Navathe, Fundamentals of Database Systems, Fourth Edition, Addison Wesley, 2004 Chapters: 1.1, 1.2, 1.3, 2.1, 2.2, 3.3, 3.4, 3.5, 4.1, 4.2, 5.1, 5.2, 6.2, 7.1, 8.4, 9.1, 10.2, 10.3, 10.4, 13.8, 14.3, 18.1, 18.2, 19.1, 23.1			
Suggested Readings: 1. Abraham Silberschatz, Henry Korth, S. Sudarshan, Database Systems Concepts, 4th Edition, McGraw Hill, 1997. 2. Jim Melton, Alan Simon, Understanding the new SQL: A complete Guide, Morgan Kaufmann Publishers, 1993.			

Course Title	Object Oriented Programming Concepts	L	T	P	C	Periods
Course Code	UCSH-402	4	0	0	4	56
Course Category	DSC					
Course Objectives: To understand and implement standard tools and techniques for software development using object oriented approach.						
Course Outcomes: Develop the skill set to 1. Analyze and design real world problems using OOPS concepts viz: Encapsulation, object identity, inheritance and polymorphism. 2. Handle exceptions and write generic programming 3. Write simple C++ programs using basic object oriented paradigms						
Prerequisites for the Course: UCSH-102						
Unit	Topic	Content				Periods
1	Introduction	Introducing Object-Oriented Approach, complexity of the structure and design of complex systems, Relating to other paradigms (functional, data decomposition). Basic terms and ideas: Abstraction, Encapsulation, Inheritance, Polymorphism, Review of C, Difference between C and C++ - cin, cout, new, delete operators.				14
2	Classes and Objects	Encapsulation, information hiding, abstract data types, Object & classes, attributes, methods, C++ class declaration, State identity and behavior of an object, Constructors and destructors, instantiation of objects, Default parameter value, objects, dynamic memory allocation, Metaclass/abstract classes.				14
3	Inheritance and Polymorphism	Inheritance, Class hierarchy, derivation – public, private & protected, Aggregation,				14

		composition versus classification hierarchies, Polymorphism, Categorization of polymorphism techniques, Method polymorphism, Polymorphism by parameter, Operator overloading, Parametric polymorphism	
4	Generic function	Template function, function name overloading, Overriding inheritance methods, Runtime polymorphism, Multiple Inheritance. Files and Exception Handling: Persistent objects, Streams and files, Namespaces, Exception handling, Generic Classes	14
		Total	56

Key Text:

1. A.R.Venugopal, Rajkumar, T. Ravishanker, Mastering C++, TMH, 1997.
Chapters: 1 (1.1.-1.18), 2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 14, 15, 16 (16.1-16.3), 17 (17.1-17.2), 18 (18.1-18.9), 19 (19.1-19.7)

Suggested Readings:

1. R. Lafore, "Object Oriented Programming using C++", Galgotia Publications, 2004.
2. James Rumbaugh "Object Oriented Modeling and Design" Prentice Hall Publications, 1991
3. D. Parsons, "Object Oriented Programming with C++", BPB Publication.
4. Steven C. Lawlor, "The Art of Programming Computer Science with C++", Vikas Publication.
5. Schildt Herbert, "C++: The Complete Reference", 4th Ed., Tata McGraw Hill, 1999.
6. Tony Gaddis, Watters, Muganda, "Object -Oriented Programming in C++", 3rd Ed., Wiley Dreamtech, 2004
7. Grady Booch Object oriented Analysis and Design with applications
8. S. B. Lippman & J. Lajoie, "C++ Primer", 3rd Edition, Addison Wesley, 2000.

Course Title	Software Lab in C++	L	T	P	C	Periods
Course Code	UCSH-403	0	0	6	3	84
Course Category	DSC					
Course Objectives: To analyze a real world problem and code using object oriented concepts of C++ .						
Course Outcomes: Develop the skill set to 1. Understand and implement object oriented paradigms viz., Encapsulation, object identity, inheritance and polymorphism. 2. Develop a mini project.						
Prerequisites for the Course: UCSH-102, UCSH-202						
Unit	Topic	Contents				Periods
1	Classes and Objects	To set and get the values given by the user, calculate the area/perimeter of geometrical shapes, Using friend functions. Static members and functions to count the number of objects created. Using constructors (default, parameterized, overloaded) defined in (local, global and function) and destructors along with their order of invocation in addition / multiplication of complex numbers (passing objects as parameters)by using inline functions. matrix multiplication using dynamic objects.				22
2	Inheritance: Single, Multiple, Hierarchical, Multilevel, Hybrid.	To write sample codes with different access specifiers, visibility modes and various types of inheritance, To use constructors and destructors in all the types of inheritance to understand the order of invocation.				12
3	Polymorphism, Templates, Exception handling,	Operator overloading on: arithmetic, logical, assignment, relational, unary, subscripting, new and delete operators, using abstract classes with virtual functions. Developing sample codes for swapping,				22

	Namespaces, Files	maximizing two numbers using templates. Handling division by zero exception, creating own namespace. To read and write into a file.	
4	Mini project	Student academic/ non-academic activities, Banking systems/ Self-reliance departments/ etc., using all the concepts.	28
		Total	84
Reference: 1. A.R. Venugopal, Rajkumar, T. Ravishanker “Mastering C++”, TMH, 1997. 2. Schildt Herbert, “C++: The Complete Reference”, 4th Ed., Tata McGraw Hill, 1999.			

Semester-V

Course Title	Operating Systems	L	T	P	C	Periods
Course Code	UCSH - 501	4	0	0	4	56
Course Category	DSC					
Course Objective: To give a basic idea of an OS and its functionalities. To give an in- depth understanding of CPU and its scheduling, memory management, and device management. This course covers other functionalities of OS such as file systems, virtual memory, disk request scheduling, concurrent processes, deadlocks, security, and integrity.						
Course Outcome: Develop the skill set to 1. To become familiar with the working of the general operating system. 2. To give perspectives to understand the OS services. 3. To know the working of the operating system processes management. 4. To identify and understand memory subsystems. 5. To know the basic file management system. 6. To understand the storage structure.						
Prerequisites for the Course: NA						
Unit	Topics	Contents				Periods
1	Introduction	What is an Operating System, Simple Batch Systems, Multiprogrammed Batches systems, Time-Sharing Systems, Personal -computer systems, Parallel systems, Distributed Systems, Real -Time Systems, Computing Environments, Computer System operation, I/O Structure, Storage structure and Hierarchy, Operating System components and Services- System calls, System Programs, System Structure				12
2	Processes	Process Concept, Process Scheduling, Operation on Processes CPU Scheduling: Basic Concepts, Scheduling Criteria, Scheduling Algorithms, Multiple-Processor Scheduling, Process Synchronization: Background, The Critical -Section Problem, Synchronization Hardware, Semaphores, Classical Problems of Synchronization				12
3	Deadlocks	System Model, Deadlock Characterization, Methods for Handling Deadlocks, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Recovery from				10

		Deadlock.	
4	Memory Management	Background, Logical versus Physical Address space, swapping, Contiguous allocation, Paging, Segmentation, Virtual Memory: Demand Paging, Page Replacement, Page-replacement Algorithms, Performance of Demand Paging, Allocation of Frames, Thrashing, Other Considerations.	12
5	Storage Management	File system Implementation, Directory Implementation, Allocation methods, free space management, Disk Structure, Disk Scheduling, Disk Management, Swap-Space Management, Disk Reliability	10
		Total	56
Reference: Silberschatz and Galvin, “Operating System Concepts”, Wiley-India, International Student Version, 8th Ed., 2010 Chapters: Text 1: 1, 3 (3.1, 3.2, 3.3), 5 (5.1, 5.2,5.3, 5.5), 6 (6.1, 6.2, 6.4, 6.5, 6.6), 7, 8 (8.1, 8.2, 8.3, 8.4, 8.6), 9 (9.2,9.4,9.5,9.6,9.9), 11(11.2-11.5), 12(12.2,12.4-12.6).			

Course Title	Computer Networks	L	T	P	C	Periods
Course Code	UCSH-502	4	0	0	4	56
Course Category	DSC					
Course Objective	1. To develop an understanding of modern network architectures from design and performance perspective. To introduce the student to the major concepts involved in wide-area networks (WANs), local area networks (LANs) and Wireless LANs (WLANs). 2. To provide an opportunity to do network programming To provide WLAN measurement ideas.					
Course Outcomes: Develop the skill set to 1. Explain the functions of the different layers of the OSI Protocol. 2. Draw the functional block diagram of wide-area networks (WANs), local area networks (LANs) and Wireless LANs (WLANs) to describe the function of each block. 3. For a given requirement (small scale) of wide-area networks (WANs), local area networks (LANs) and Wireless LANs (WLANs) design it based on the market available component 4. For a given problem related TCP/IP protocol developed the network programming. 5. Configure DNS DDNS, TELNET, EMAIL, File Transfer Protocol (FTP), WWW, HTTP, SNMP, Bluetooth, Firewalls using open source available software and tools.						
Prerequisites for the Course: UCSH-401						
Unit	Topic	Contents				Periods
1	Introduction	Introduction to computer networks, evolution of computer networks and its uses, Advantages and Disadvantages of Computer Network, reference models: OSI reference Models, TCP/IP Protocol Suite Networking fundamentals: Internet, Circuit switching vs Packet switching, ISPs, Delay and Loss in Packet Switched Networks				10
2	Local Area Network	LAN Architecture, LAN topologies- Bus/ Tree LAN, Ring LAN, Star LAN, Wireless LAN, Ethernet and Fast Ethernet, Token Ring				8
3	Application layer and data link layer	Application Layer Protocols: HTTP, FTP, SMTP, DNS Data link layer design issues, Flow Control- Stop and Wait, Error Detection, Error Control, error detection and correction, data link layer protocols,				10

		sliding window protocols, example of data link protocol- HDLC	
4	Medium access layer	Channel allocation problem, multiple access protocols, Introduction to ALOHA, CSMA/CD, CSMA/CA	8
5	The network layer	Introduction, Routers, Network layer concepts, shortest path routing, flooding, distance vector routing, link state routing (without algorithms), congestion control and quality of service, internetworking, IP, Ipv4 Addressing vs Ipv6	10
6	The transport layer	The transport layer services, elements of transport protocols, TCP and UDP, Brief introduction to presentation and session layer, E-mail	10
		Total	56
Reference: 1. Data Communication & networking: Forouzan, B. A. , Vth Ed, The McGraw-Hill Companies, Inc - 2017. 2. Data And Computer Communications, by William Stallings, VII edition, Pearson Education, 2005.			
Suggested Readings: 1. Andrew S. Tanenbaum, Computer Networks, Fourth Edition, Pearson Education, 2003 2. Jim Kurose, Keith Ross, A Top Down Approach Featuring the Internet , Third Edition, Pearson Education, 2004			

Course Title	Software Lab in Java	L	T	P	C	Periods
Course Code	UCSH-503	0	0	6	3	84
Course Category	LW					
Course Objectives: This course introduces the basics of Java programming.						
Course Outcome: Develop the skill set to 1. Write java code for basic programming. 2. Understand and implement object oriented concepts like Abstraction, Encapsulation and Inheritance 3. Implement various data structures like List, Tree and Graph 4. Understand related concepts in Java like Exceptions, Packages 5. Implement simple programs with graphics in Java						
Prerequisites for the Course: NA						
Unit	Topic	Contents				Periods
1	Primitives and Objects in Java	Simple programs for learning primitive data, basics of creating, using objects in java				10
2	Loops and Control structures	Simple programs using if-else, for, while, switch-case, break, continue				10
3	Access modifiers and Constructors	Private, default, protected and public access modifiers for fields and methods. Default Constructor, Constructor invocations, static, abstract, final				15
4	Inheritance	Object class, super, Multi-level inheritance, Typecasting, Constructors in Inheritance, Fields and Methods in Inheritance, Overriding, Final & Abstract classes				15
5	Arrays, Exceptions, Packages, Interfaces	1D and 2D arrays in Java, Inheritance in Arrays, Checked and Unchecked Exceptions, try, catch, throw, finally, Different ways to import packages, Class path, Inheritance in Packages, Using interfaces for multiple inheritance				14
6	Graphic, sets, and Process	Swing, Collection, Multithreading				20

		Total	84
Reference & Suggested Readings: 1. Herbert Schildt, Java: The Complete Reference, , McGraw-Hill Education, 12th edition, 2021, ISBN-13: 978-1260463415. 2. Cay S. Horstmann, Core Java Volume I – Fundamentals, 12th Edition, Pearson Education, 2021 ISBN-13: 978-0137673620 3. Kathy Sierra & Bert Bates, Head First Java, O'Reilly Media, 2005, ISBN-13: 978-059600920			

Semester-VI

Course Title	Software Engineering	L				Periods
Course Code	UCSH - 602	4				56
Course Category	DSC					
Course Objectives: This course introduces students to the general approach of building a software product. It also introduces the various methods of building software products. It also covers the OO approach of software design and other phases.						
Course Outcome: Develop skill set to 1. Design a software model for a software product 2. Estimate the cost of planning and implementing a software product 3. Calculate the software metrics at different phases of development						
Prerequisites for the Course: NA						
Unit	Topic	Contents				Periods
1	Introduction and a Generic View of Software Engineering Processes	The Nature of Software, Defining the Discipline, The Software Process, Software Engineering Practice and How it all Starts; Process Models, Agility and Process, and Recommended Process Models				10
2	Human Aspects and Practice Principles	Human Aspects of Software Engineering: Characteristics of Software Engineer, Psychology of Software Engineering, Software Team, team Structures, Social media and Global teams; Practice Principles: Core Principles, Principles that guide each framework activity				10
3	Requirement Analysis and Modeling	Requirement Analysis: Requirements Engineering, Establishing the GroundWork, Requirements Gathering, Developing the Use cases, building the Analysis Model, Negotiating Requirements, requirements Monitoring, Validating requirements; Requirement Modeling: requirement Analysis, Scenario Based Modeling, Class based modeling, functional				12

		Modeling, Behavioral Modeling	
4	Design and Architecture	Design Concepts: Design within the context of SE, Design Process, Design Concepts, Design Model; Architecture Design: Software Architecture, Agility and Architecture, Architectural Styles, Architectural Considerations, Architectural Decisions, Architectural Design, Assessing Architectural Design; Component Level Design: What is Component, Designing Class based Component, Conducting Component Level Design, Component Factoring	12
5	Quality, Security and Testing	Quality: What is Quality?, Software Quality, Software Quality Dilemma, Archiving Software Quality; Security: Security Information, Life-cycle Models, Development Life Cycle Models, Requirements Engineering, Mis-use and Abuse cases and Attack, Risk Analysis, Threat Modeling, Prioritization, and Mitigation, Attack Surface, and Secure coding; Testing-Component level: Strategic Approach, Planning and Record Keeping, test case Design, White-Box Testing, Black-Box Testing; Testing-Integration Level: Fundamentals, Integration Testing, AI and Regression testing, Validation Testing, Testing Patterns;	12
		Total	56

Reference:

Roger S Pressman and Bruce R. Maxim, Software Engineering: A Practitioner's Approach , Ninth Edition, McGraw Hill

Suggested Readings:

1. Somerville, Software Engineering, Ninth Edition, Addison-Wesley, 2011
2. Pankaj Jalote, An Integrated Approach To Software Engineering, Third Edition, Narosa Publishing House, New Delhi, 2005

Course Title	Design and Analysis of Algorithms	C	L	T	P	Periods
Course Code	UCSH-603	4	4	0	0	56
Course Category	DSC					
Course objective: • To develop problem solving skills by analyzing various problems and learning the techniques for implementation. • Analyze the asymptotic performance of algorithms. • Write rigorous correctness proofs for algorithms. • Demonstrate a familiarity with major algorithms and data structures. • Apply important algorithmic design paradigms and methods of analysis. • Synthesize efficient algorithms in common engineering design situations.						
Course outcome: Develop the skill to 1. Analyze the problem. 2. Implement various problem solving methods. 3. Write algorithms for implementing solutions. 4. Discriminate between different problem solving approaches. 5. For a given algorithm, analyze worst-case running times of algorithms based on asymptotic analysis and justify the correctness of algorithms						
Prerequisites for the course: UCSH-101, UCSH-201						
Unit	Title	Contents				Periods
1	Introduction	Algorithm, Algorithm Specification, and Performance Analysis. Randomized Algorithms. Basic Data Structures: Stacks and Queues, Trees, Dictionaries, Priority Queues, Sets and disjoint Set Union, Graphs.				8
2	Divide and Conquer, Greedy methods	Divide and Conquer: General method, Binary search, Finding Maximum and Minimum, Merge sort, Quick sort, Selection, Strassen’s Matrix Multiplication; Greedy: General Method, Knapsack problem, Job Sequencing with deadlines, Minimum cost spanning Trees, single source shortest path.				6

3	Dynamic Programming	General Method, All pairs shortest path, single source shortest path, Optimal Binary Search Trees, 0/1 Knapsack, reliability design, the traveling salesperson problem.	8
4	Traversal, Searching and Back tracking Techniques.	Traversal, searching: Techniques for Binary Trees, Graphs (BFS, DFS), connected components spanning trees, Biconnected components; Back Tracking: General Method, 8-queen's problem, sum of subsets, Graph colouring, Hamiltonian cycles, Knapsack problem.	8
5	Branch and Bound, NP-Hard and NP-Complete Problems.	Branch and Bound: The general method Least cost Search, Bounding, FIFO Branch and Bound, LC Branch and Bound, 0/1 Knapsack problem: LC and FIFO Branch and Bound Solutions; NP-Hard and NP-Complete: Basic concepts, Nondeterministic algorithms. The classes NP- hard and NP-complete. Cook's theorem; NP-Hard graph problems: CDP, NCDP, CNDP, Code generation with common subexpressions, Some simplified NP-hard Problems.	6
		Total	56

Key Text:

E Horowitz, S Sahani S Rajasekaran, Fundamentals of Computer Algorithms, Second Edition, Universities Press, 2008.

Suggested Texts:

1. Alfred V. Aho and John E. Hopcraft, and Jeffrey D. Ullman, The Design Analysis of Computer Algorithms, Pearson, 1974
2. Thomas H. Cormen, Charles E. Leiserson, R.L. Rivest, Introduction to Algorithms, Prentice Hall of India Publications, New-Delhi, 2008
3. Sara Baase and Allen Van Gelder, Computer Algorithms: Introduction to Design and Analysis, Third Edition, Pearson Education (Singapore) Pvt. Ltd, New Delhi, 2000
4. Alfred V. Aho, John E. Hopcroft, Jeffrey D. Ullman, The Design and Analysis of Computer Algorithms, Pearson Education (Singapore) Pvt. Ltd New Delhi. 2012

Course Title	Software Lab in Web Programming	L	T		C	Periods
Course Code	UCSH-604	0	0		3	84
Course Category	DSC					
Course Objectives: This course introduces various methods of developing web applications						
Course Outcome: Develop the skill set to 1. Create web pages using HTML code; 2. Create web applications for client/server communication 3. Create web server applications using database connectivity.						
Course Prerequisites for the Course: NA						
Unit	Topic	Contents				Periods
1	HTML	HTML pages using all tags; Tables and Forms				18
2	CSS	Box model; various styling options				12
3	Java Scripts	script basics; functions; objects; control statements; event handling; DOM; event handling; AJAX				18
4	Servlets	Servlet creation; request/response methods; session management; database connectivity				18
5	JSP	creating server pages with different tags; database connectivity; MVC model				18
		Total				84
References: 1. Web Technologies - Theory and Practice - M Srinivasan Pearson, 2012 Web resources						
Suggested Readings: 1. Jon Duckett, HTML and CSS: Design and Build Websites, Wiley, 1st Edition, 2011, ISBN-13: 978-1118008188 2. Jennifer Niederst Robbins, Learning Web Design: A Beginner’s Guide to HTML, CSS, JavaScript, and Web Graphics, 5th Edition, O’Reilly Media, 2018, ISBN-13: 978-1491960202						

Semester-VII

Course Title	Linux System Programming					Periods
Course Code	UCSH-703					56
Course Category	DSC					
Course Objectives: The main objectives are: 1. To learn the different set of system calls for the Linux Operating System 2. To understand how the Linux OS manages files, processes and memory 3. To implement inter-process communication using different mechanisms						
Course Outcomes: To develop skill set to: 1. Assimilate the internal abstractions of any Operating System 2. Utilize the insights gained from how these abstractions were implemented and apply them in other areas of work						
Prerequisites for the course: UCSH-102, UCSH-201, UCSH-202, UCSH-401,						
Unit	Topic	Contents				Periods
1	Introduction	System Calls, Library Functions, Standard C Library, Error handling.				6
2	File Management	Overview, File Operations (open, read, write, lseek, close), Atomicity, File Descriptors - relation to open files and duplication, File I/O variations (pread, pwrite, readv, writev), File truncation				9
3	Process Management	Process concept, Process Memory Layout, Virtual Memory Management, Stack Frames, Command line arguments, Environment Variables, Process - Creation, Termination, Execution and Monitoring				14
4	Memory Management	Heap and Stack Memory allocation, Memory Mapping - Creation, Unmapping, File mapping, Synchronization, Anonymous Mapping				14
5	Inter-Process Communication	Signals, Pipes, FIFO, POSIX Semaphores - Named Semaphore and Semaphore Operations, POSIX Shared Memory - Creation, Usage & Removal				13
		Total				56

Reference: Michael Kerrisk, The Linux Programming Interface, No Starch Press, 2010.
Chapters: 3 (3.1 - 3.4), 4 (4.1 - 4.7), 5 (5.1 - 5.8), 6 (6.1 - 6.7), 7, 20, 24, 25, 26, 27, 44 (44.1 - 44.4, 44.6 - 44.8), 49 (49.1 - 49.5, 49.7), 53 (53.1 - 53.3), 54 (54.1 - 54.4)

Suggested Readings:

1. Robert Love, Linux System Programming, O'Reilly, 2013
2. W. Richard Stevens and Stephen A. Rago, Advanced Programming in the UNIX environment, Addison-Wesley, 1992

Semester-VIII

Course Title	Formal Languages	L	T	P	C	Periods
Course Code	UCSH- 802	4	0	0	4	56
Course Category	DSC					
Course Objectives: 1. to learn fundamental theory of computer science 2. to learn formal methods of computation like automata theory, formal languages & grammars, and Turing machines.						
Course Outcomes: Develop skill set to: 1. understand basic properties of formal languages and grammars 2. differentiate between regular, context-free and recursively enumerable languages 3. generalize the statements of a language to construct a grammar 4. generate statements given the grammar of a language 5. classify machines by their power to recognize languages						
Prerequisites: UCSH-201, UCSH-603						
Unit No.	Title	Contents				No. of Periods
1	UNIT I	Deterministic Finite Automata, Non-deterministic automata, Equivalence of NFA and DFA, Reduction in number of states in Finite Automata				9
2	UNIT II	Regular Languages and Regular Grammars: Regular expressions, Connection between regular expressions and regular languages, Regular Grammars, Closure properties of Regular languages, Pumping lemma-Non regular languages.				9
3	UNIT III	Context free grammars ,Parsing Ambiguity, Context free grammars and programming languages ,Simplification of complex grammars and their normal forms(Chomsky and Greibach Normal form.				9
4	UNIT IV	Pushdown Automata: Nondeterministic pushdown automata, deterministic pushdown automata, Pushdown automata and context free languages, Context free grammars and pushdown automata), Properties of context				9

		free languages, Pumping lemma for non-context free languages	
5	UNIT V	Hierarchy of formal languages and automata: Recursive languages and Recursively enumerable languages, Unrestricted grammars, Context sensitive grammars and languages, Chomsky hierarchy	10
6	UNIT VI	Turing machines: Standard Turing machine, Combining Turing machine for complicated tasks, Turing thesis, Variations in Turing machine, Linear Bound automata.	10
		Total	56
Key Text: Introduction To Formal Languages And Automata, by P.Linz, Narosa Pub. 1997. REFERENCES 1. Formal Languages, by A.Salomaa , Academic Pub.1973. 2. Introduction To Formal Languages , by Gyorgy E Revesz, Mcgraw-Hill Book Co.,1985			

Specialization Courses

Artificial Intelligence:

1. UCSH-AI-Artificial Intelligence
2. UCSH-DL-Deep Learning
3. UCSH-MLO-Machine Learning Operations
4. UCSH-NLP-Natural Language Processing

Course Title	Artificial Intelligence	L	T	P	C	Periods
Course Code	UCSH-AI	4	0	0	4	56
Course Category	DSE					
Course Objectives: ● To understand an autonomous agent ● To study a wide variety of search methods that agents can employ for problem solving.						
Course Outcomes: Students will be able to 1. Understand a problem from autonomous agent’s point of view 2. Solve a real-world problem using various search methods to arrive at a human like decision 3. Write inference rules based on logic that can make decisions.						
Prerequisites: Discrete Mathematics, Problem Solving with Computer, Data Structures and Algorithms						
Unit	Topic	Contents				Periods
1	Introduction	What is AI? The foundations of AI, The history of AI, The state of the art; Intelligent Agents: Agents and Environments, The Concept of rationality, Nature of Environment and the structure of Agent ;				10
2	Problem Solving Problems by Searching	Solving Problems by searching: Problem solving agents, Searching for solutions, Informed and Uninformed Search strategies, Heuristic Functions;				10
3	Adversarial Search	Games, Optimal Decisions in games, Alpha, beta pruning, Imperfect Real- Time Decisions; Constraint Satisfaction Problems: Defining CSP, Constraint Propagation, Backtracking and Local Search for CSP,				12

		The Structure of Problem;	
4	Knowledge, Reasoning and Planning	Logical Agents: Knowledge-Based Agents, The Wumpus World, Logic, Propositional Logic, Propositional Theorem Proving, Effective Propositional Model Checking, Effective Agent Based on Propositional Logic; Inference in the First Order Logic: Propositional vs First Order Inference, Unification and Lifting, Forward chaining, Backward Chaining, Resolution	12
5	Classical Planning	Definition, Algorithm for Planning as State space search; Planning Graphs, Other Classical Planning approaches, Analysis of Planning Approaches; Planning and Acting in the Real World: Time, Schedules and Resources, Hierarchical Planning, Planning and Acting in Non-Deterministic Domains, Multi Agent Planning;	12
		Total	56
Text: Stuart Russell, Peter Norvig, Artificial Intelligence: A Modern Approach, Third Edition, Pearson Education, 2011 References: 1. Elaine Rich, Kevin Knight and Shiv Shankar B. Nair, Artificial Intelligence, Third Edition, Tata McGraw Hill, 2009 2. Deepak Khemani, A First Course in Artificial Intelligence, McGraw Hill Education (India), 2013			

Course Title	Deep Learning	L	T	P	C	Periods
Course Code	UCSH-DL	3	0	2	4	70
Course Category	DSE					
Course Objectives: Introduce the main deep learning architectures used in the industry. Develop a strong understanding of the training mechanisms. Expose students to Advanced topics like Attention based models, Deep Generative models						
Course Outcomes: Through this course, students will be able to: 1. Develop deep learning solutions to any AI related problem 2. Apply different optimization techniques to train neural networks 3. Apply different dimensional reduction techniques using neural networks 4. Develop and train a CNN model to solve a real world problem based on Image data. 5. Develop and train a RNN model to solve a real world problem based on sequential data 6. Have a strong understanding of the internal workings of advanced deep learning architectures like VAEs and GANs 7. Implement several deep learning algorithms from scratch 8. Prototype solutions in Python						
Prerequisites: Machine Learning, Optimization						
Unit	Title	Contents				Periods
1	An Introduction to Neural Networks	McCulloch Pitts Neuron, Perceptrons, Perceptron Learning Algorithm and Convergence, Multilayer Perceptron (MLPs), Representation Power of MLPs, Sigmoid Neurons, Gradient Descent, FeedForward Neural Networks, Backpropagation.				8
2	Gradient Descent Strategies	Gradient Descent (GD), Momentum Based GD, Nesterov Accelerated GD, Stochastic GD, AdaGrad, RMSProp and Adams				5
3	Embedding and Representation Learning	Review of Principal Component Analysis, Autoencoders and relation to PCA, Regularization in autoencoders, Denoising autoencoders, Sparse autoencoders, Contractive autoencoders				5
4	Regularization for Deep Neural Networks	Bias variance Tradeoff, $\square^2$ - regularization, Early stopping, Dataset augmentation, Parameters sharing and tying, Injecting noise at input, Ensemble methods, Dropout, Greedy Layer Wise Pre-training, Better				6

		activation functions, Better weight initialization methods, Batch Normalization	
5	Convolutional Neural Networks	The basic structure of convolutional Network: Padding, Strides, ReLU Layer, Pooling, and Fully Connected Layers; Case studies of Convolutional Architectures: AlexNet, ZFNet, VGGNet, GoogLeNet, ResNet; Visualizing Convolutional Neural Networks, Guided Backpropagation.	6
6	Recurrent Neural Networks	Recurrent Neural Networks, Backpropagation through time (BPTT), Vanishing and Exploding Gradients, Truncated BPTT, GRU, LSTMs	6
7	Advanced topics in Deep Learning	Encoder Decoder Models, Attention Mechanism, Attention over images, Variational autoencoders, Generative Adversarial Networks (GANs)	6
Key Text(s): 1. Ian Goodfellow, Yoshua Bengio, Aaron Courville, Deep Learning, The MIT Press, 2016 2. Nikhil Buduma, Nicholas Locascio, Fundamentals of Deep Learning: Designing Next-Generation Machine Intelligence Algorithms, O'Reilly Media, 2017 3. Charu C. Agarwal, Neural Networks and Deep Learning: A Textbook, Springer, 2018			
References: 1. Duda, R.O., Hart, P.E., and Stork, D.G, Pattern Classification, Wiley-Interscience, Second Edition, 2001 2. Theodoridis, S. and Koutroumbas, K., Pattern Recognition. Fourth Edition, Academic Press, 2008 3. Russell, S. and Norvig, N. Artificial Intelligence: A Modern Approach, Prentice Hall Series in Artificial Intelligence. 2003 4. Bishop, C. M. Neural Networks for Pattern Recognition. Oxford University Press. 1995. 5. Hastie, T., Tibshirani, R. and Friedman, J. The Elements of Statistical Learning. Springer. 2001. 6. Koller, D. and Friedman, N. Probabilistic Graphical Models. MIT Press. 2009.			
UCSH-DL: Practicals - Deep Learning			
Unit	Title	Contents	Periods
1	PyTorch Basics	Why PyTorch?, Basics of Tensors: Tensors, indexing, size, offset, slicing, dtypes, moving tensors to GPU	10

		Representation of real-world data using tensors: image data, tabular data, and text data Download datasets, Dataset class, Dataset Transformation, Normalizing the data, Create your own dataset using <i>dataloader</i>	
2	Working with pretrained models	Pretrained models: AlexNet, ResNet, Artistic Style, CycleGAN etc	4
3	The mechanics of Learning	Gathering some data, Visualizing the data, Iterating to fit the model, Normalizing inputs, pytorch autograd, pytorch.nn module	6
4	Case studies	1. Classification of images in CIFAR10 dataset 2. Classification model to detect suspected tumors Sentiment Analysis	8
		Total	28
Key Text(s): 1. Eli Stevens, Luca Antiga, and Thomas Viehmann, Deep Learning with PyTorch, Manning Publications Co., 2020 2. PyTorch documentation, Retrieved July 05, 2022 from https://pytorch.org/docs/stable/index.html PyTorch For Deep Learning, freecodecamp.org, retrieved July 05, 2022 from https://www.youtube.com/watch?v=GIsg-ZUy0MY			

Course Title	Machine Learning Operations					Periods
Course Code	UCSH-MLO	3	0	2	4	70
Course Category	DSE					
Course Objectives: Maintenance of the Machine learning models in the production environment has become one the challenges for modern organizations using technology as well as business. This course is designed to have a basic understanding of the ML model life cycle in the production environment.						
Course Outcomes: Develop the skill set to 1. Understand the MLOps as a discipline in the industry 2. Learn the ML- life cycle 3. Example based learning on ML models 4. Web Applications for ML Models 5. Containerization						
Prerequisites for the Course: Machine Learning, Python Programming						
Unit	Title	Contents				Periods
1	MLOps: What and Why	Why Now and Challenges, People of MLOps, Key MLOps Features.				12
2	MLOps: How	Developing Models, Preparing for Production, Deploying to Production, Monitoring and Feedback Loop, Model Governance.				15
3	MLOps: Examples	Consumer Credit Risk Management, Marketing Recommendation Engines,Consumption Forecast				15
		Total				42
Key Text(s): Mark Trveil and the Dataiku Team, Introducing MLOps: How to Scale Machine Learning in the Enterprise, O'Reilly, 2020						

Retrieved Jun 30, 2022 from <https://itlligenze.com/uploads/5/137039/files/oreilly-ml-ops.pdf>

UCSH-MLO: Practicals - Machine Learning Operations

Unit	Title	Contents	Periods
1	Review of ML Models	Basic Scripts-PYTHON-Model Zoo	6
2	Web Application	Converting the ML into a Web App using Flask	6
3	Containerization	Docker and Docker swarm	8
4	Scaling & Orchestration	Docker and Travis CI, AWS, and Google Kubernetes	8
		Total	28

Key Text:

Sandeep Giri, MLOps - Complete Hands-On Guide with Case Study, Article, August 23, 2021.

Retrieved July 5, 2022 from <https://cloudxlab.com/blog/mlops-machine-learning-operations-a-complete-hands-on-guide-with-case-study/>

Course Title	Natural Language Processing	L	T	P	C	Periods
Course Code	UCSH-NLP	3	0	2	4	70
Course Category	DSE					
Course Objective: To impart knowledge in classification of documents, retrieving and extracting information from documents, identifying important documents available as unstructured text.						
Course Outcome: Develop the skill set to: 1. Understand basic elements of NLLP 2. Apply n-gram techniques for text processing/ text Classification 3. Apply Machine Learning models for Text Classification 4. Develop Sequence Models and Language Models for real-world problems.						
Prerequisites: Machine Learning, Deep Learning						
Unit	Title	Contents				Periods
1	Introduction to NLP	Regular Expressions, Words, Corpora, Text Normalization, Edit Distance				4
2	N-gram Language Models	N-grams, Evaluating Language Models, Sampling sentences, Generalization, Smoothing, Entropy & Perplexity				4
3	Text Classification	Naive Bayes classifier, Optimizing for sentiment analysis, Naive Bayes Language Model, Naive Bayes for other text classification tasks, Evaluation, Cross-validation, Logistic Regression, Multinomial Logistic Regression, Cross-entropy loss, Gradient Descent, Regularization				7
4	Vector Semantics	Lexical Semantics, Vector Semantics, Words & Vectors, Cosine Distance, TF-IDF, Pointwise Mutual Information, Word2Vec, Visualizing embeddings, Semantic properties of embeddings, Evaluation vector models				6
5	Sequence Labeling	Part-of-Speech Tagging, Named Entity Tagging, HMM PoS tagging, Conditional Random Fields, Evaluation				5

6	Sequence Processing using Deep Learning	Feedforward Networks for NLP, Neural Language Models, RNN Language Models, Stacked and Bidirectional RNNs, LSTM, Transformers, Contextual generation and Summarization	8
7	Machine Translation and Encoder-Decoder Models	Language Divergence, Encoder-Decoder Model, Encoder-Decoder with RNN, Attention, Beam Search, Encoder-Decoder with Transformers, Bidirectional Transformer Encoders, Transfer Learning, Evaluation	8
Total			42
Key Text(s): Dan Jurafsky and James Martin, An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition, Speech and Language Processing, Third Edition, Prentice Hall, 2022.			
References: 1. Jacob Eisenstein, Introduction to Natural Language Processing, MIT Press, 2019 2. Steven Bird, Ewan Klein and Edward Loper, Natural Language Processing with Python - Analyzing Text with the Natural Language Toolkit, O'reilly First edition, 2011			
UCSH-NLP: Practicals - Natural Language Processing*			
Unit	Title	Contents	Periods
1	Regular Expressions	Implement Regular expression parsing in Python	4
2	Byte Pair Encoding	Implement Byte Pair Encoding in Python	2
3	Edit Distance	Implement minimum edit distance and alignment for spelling correction	4
4	PoS Tagging	Implement Parts-of-Speech tagging	4

5	Word Vectors	Implement Word Vectors using existing python libraries and Implement Word2Vec from scratch	2
6	Neural Networks	Implement forward and back propogation from scratch, Implement networks using matrices	6
7	Encoder-Decoder model	Implement Encoder-Decoder model with RNN	2
8	Transformer	Implement a Transformer model from scratch	4
		Total	28
*All implementation using PyTorch/TensorFlow/JAX			
1. Delip Rao and Brian McMahan, Natural Language Processing with PyTorch - Build Intelligent Language Applications Using Deep Learning, O'Reilly, First Edition. 2019. 2. Sebastian Raschka, Build a Large Language Model (From Scratch), Manning, 2024.			

**SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING
(DEEMED TO BE UNIVERSITY)**



**Syllabus for Minor Disciplines offered by
Department of Mathematics and Computer Science**

PRASANTHI NILAYAM- 515134
Sri Sathya Sai District, Andhra Pradesh
Tel: 08555- 287239
www.sssihl.edu.in

(EFFECTIVE FROM JUNE 2024-2025 ONWARDS)

SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING
(Deemed to be University)

DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE

Syllabus for Minor Disciplines Offered for Undergraduate Programs
(Effective from 2024-2025 onwards)

INTRODUCTION

Department of Mathematics and Computer Science (DMACS) offers minor courses in the following Disciplines for undergraduate students with required eligibility:

1. Mathematics
2. Actuarial Science
3. Data Science

These Minor degrees or courses will be offered subject to available logistics such as faculty, timetable slots etc. Proper understanding must be sought from the course coordinator/HoD before choosing to do a minor course in a particular semester.

Minor Certification: A candidate who chooses to do a minor has to earn a minimum of 12 credits in 3 years of B.S program or 16 credits in 4 years of B.S Hons/B.S Hons with Research program. It is to be noted that, the minimum credits mentioned above must be earned from the courses which are of difficulty level 300 or above. A student majoring in a discipline cannot obtain a minor in the same discipline.

MOOCs Courses for Minor Credit Transfer: A candidate can choose to earn up to 12 minor credits from online MOOCs Courses.

1. Mathematics Minor

Prerequisite: 12th standard from any stream with Mathematics as one of the core subjects
The following courses are offered for Mathematics as minor.

100 Differential Calculus
100 Integral Calculus
200 Linear Algebra -I/300 Complex Analysis
200 Algebraic Structures I/300 Methods of Differential Equations
300 Optimization Techniques
300 Numerical Analysis
400 Algebraic Structures -II
400 Mathematical Modelling

Note: For detailed syllabus, consult the syllabus document for the program BSc Mathematics/BSc(Hons) Mathematics/BSc(Hons) Mathematics with Research.

2. Actuarial Science Minor

Prerequisite: 12th standard from any stream with Mathematics as one of the core subjects

The following list of courses offered for Computer Science as minor

- 101 Actuarial Probability and Statistics
- 102 Actuarial Mathematics
- 301 Actuarial Applied Statistical Methods
- 302 Business Economics with Actuarial Applications
- 303 Business Accounting & Finance with Actuarial Applications
- 401 Regulation and Financial reporting with Actuarial Application

Note: For a detailed syllabus, consult the syllabus document for the program BSc Mathematics/BSc(Hons) Mathematics/BSc(Hons) Mathematics with Research.

3. Data Science Minor

Prerequisite: 12th standard from any stream with Mathematics as one of the core subjects

The following list of courses offered for Data Science as minor

- UDSC-101: Calculus
- UDSC-201: Probability and Statistics
- UDSC-301: Statistics for Data Science
- UDSC-401: Linear Algebra for Data Science
- UDSC-402: Data Analysis and Visualization
- UDSC-501: Optimization for Machine Learning
- UDSC-502: Data Mining and Machine Learning
- UDSC-601: Big Data Analytics
- UDSC-701: Cloud Computing

Data Science Minor Courses Syllabus

Course Title	CALCULUS	L	T	P	C	Periods
Course Code	UDSC-101	3	1	0	4	56
Course Category	IDM					
Course Objectives: Calculus provides the tools and methods for analyzing and solving problems of change and motion quantitatively. It introduces mathematical operations of differentiation and integration and develops concepts of central importance in mathematical sciences.						
Course Outcomes: 1. Develop the skill set to compute the slope of tangent at a point for given curve 2. write a differentiable function into a series 3. find the maxima and minima of function in both one and two variables 4. compute the area under the curve for a given surface 5. perform different operators on vectors like gradient, divergence and curl 6. compute the volume of a given solid.						
Prerequisites for the Course: 10 + 2 Mathematics						
Unit	Topic	Contents				Periods
1	Preliminaries	Definitions and Examples of Types of sets, real-valued functions, domain, range of functions, continuous and differentiable functions, neighbourhood, intervals, limit points, boundary points, interior points, open set, closed set. Introduction to Sequences-Limit, convergence, divergence, oscillation. Introduction to Series - Convergence, divergence and oscillation, limit comparison and D’Alembert’s ratio tests.				10
2	Differential Calculus	Successive Differentiation: Higher-order derivatives, Leibnitz theorem. Functions of two or more variables, partial derivatives, Euler’s theorem on homogeneous functions, Total derivative, Errors and approximation, Jacobians and properties. Expansion of two variable functions using Taylor’s series. Maxima and Minima of functions of two or more variables, - Lagrange’s method of undetermined multipliers.				14
3	Integral Calculus	Multiple Integrals: Double and triple integrals, Change of order of Integration, Transformation of integrals from Cartesian to polar coordinates. Applications: Calculating area, volume, surface area using integrals. Beta and Gamma Functions: Definitions, properties, and relation between Beta and Gamma functions.				14
4	Solid Geometry	Equation of plane, Straight line, Intersection of lines, Shortest distance between two lines, Intersection of planes, Sphere, Equation of tangent plane, Quadratic surfaces.				10

5	Vector Calculus	Differentiation of vectors, Curves in space, Vector operator del, Gradient, Divergence and Curl.	8
		Total	56
Key Text: Dr. B.S.Grewal, Higher Engineering Mathematics, 44th Edition, Khanna Publications, 2021, Chapters 3(3.11 to 3.18, 3.21), 4(4.1 to 4.2), 5 (5.1-5.5,5.7,5.9- 5.12), 7 (7.1 to 7.7, 7.14 to 7.16), 8 (8.1 to 8.2, 8.4 to 8.9) 9(9.1-9.6,9.8,9.9) References: 1. Calculus, George B. Thomas, et al, 12 th Edition, 2009 2. Analytical Solid Geometry by Shanti Narayana, Dr.P.K. Mittal S Chand Publishing 3. Calculus and Analytical Geometry by G.B. Thomas and R.L. Finney.			

Course Title	Probability and Statistics	L	T	P	C	Periods
Course Code	UDSC-201	4	0	0	4	56
Course Category	IDM					
Course Objectives: Probability and Statistics is the branch of mathematics that deals with modeling uncertainty. It is important because of its direct application in areas such as computer science, finance, and telecommunications. It forms the fundamental basis for many other areas in the mathematical sciences including modern optimization methods and risk modeling. This course provides students with a formal treatment of probability and statistics.						
Course Outcomes 1. Develop the skill set to solve real world problems based on axioms of probability, conditional probability and Bayes’ theorem 2. understand the several discrete and continuous random variables 3. compute covariance and correlation of two or more than two dimensional data identify statistical distributions by observing real data sets 4. compute the moment-generating function for a given random variable 5. plot histograms, Box-plot for a given univariate data 6. plot scatter diagram for a bi-variate data						
Prerequisites for the Course: This course can be taken after UDSC-101 Calculus						
Unit	Topic	Contents				Periods
1	Elements of Probability	Sample Space, Events, counting Techniques, Axioms of Probability, Addition Rules, Multiplication Rules, Conditional Probability, Independence, Bayes’ Formula and Random Variables				12
2	Discrete Random Variables	Probability Distributions and Probability Mass Functions, Cumulative Distribution Functions, Mean and Variance of a Discrete Random variable Special Discrete Random variables: Uniform, Binomial, Geometric, Negative Binomial, Hypergeometric and Poisson Distributions				12
3	Continuous Random Variables	Probability Distributions and Probability Mass Functions, Cumulative Distribution Functions, Mean and Variance of a Discrete Random variable Special Continuous Random variables: Uniform, Exponential, Normal, Erlang, Gamma, Weibull, Lognormal and Beta Distributions. Normal Approximation to the Binomial and Poisson Distributions				14

4	Joint probability Distributions	Joint distribution of Two and more than two Random Variables, Conditional Probability Distributions and Independence, Covariance and Correlation, Common Joint Distributions, Linear Functions and General Functions of Random Variables, Moment-Generating Functions	12
5	Descriptive Statistics	Numerical Summaries of Data, Stem-and-Leaf Diagrams, Frequency Distributions and Histograms, Box Plots, Scatter Diagrams, Probability Plots.	6
		Total	56

Key Text:

Montgomery Douglas C., Runger George C., Applied Statistics and Probability for Engineers, Seventh Edition, Wiley, 2018, Chapters: 2 - 6.

Suggested Readings:

1. Sheldon M Ross, Introduction to Probability and Statistics for Engineers and Scientists, 6th Edition, Elsevier Academic Press, 2020
2. Irwin Miller Marylees Miller, John E. Freund's Mathematical Statistics with Applications, 8th Edition, Pearson Publications

Course Title	Statistics for Data Science	L	T	P	C	Periods
Course Code	UDSC-301	3	1	0	4	56
Course Category	IDM					
Course Objectives: The main purpose of Statistical Inference is to make an inference about a population based on information contained in a random sample selected from that population, and to provide an associated measure of goodness for the inference. Another purpose is hypothesis testing for mean and variance of a normal distribution.						
Course Outcomes: 1. Develop the skill set to Estimate both point and interval estimations for a given parameters based on a sample from a normal population 2. Analyze the estimated parameter 3. Find an estimator for population proportion based on sample data 4. Develop testing hypothesis for a given real world problem 5. Apply different statistical tests based on data 6. Model the relationship between a scalar response and one or more explanatory variables 7. Find an appropriate regression model for a given data						
Prerequisites for the Course: Probability and Statistics, Calculus						
Unit	Topic	Contents				Periods
1	Point Estimation	Sampling Distributions and the Central Limit Theorem, Point Estimation: Unbiased Estimators, Variance of a Point Estimator, and Mean squared Error of an Estimator, and Methods of Point Estimation: Method of Moments, Maximum Likelihood Estimators, Bayesian Estimation of Parameters;				10
2	Interval Estimation for a Single Sample	Confidence Intervals on mean of a Normal Distribution, Confidence Interval on variance of a Normal Distribution, Large Sample Confidence Interval for a Population Proportion, Bootstrap Confidence Interval, Tolerance and Prediction Intervals				9
3	Testing Hypotheses for a Single Sample	Hypothesis Testing, Tests on the Mean and variance of a Normal Distribution, Tests on a Population Proportion, Testing for Goodness of Fit and Contingency Table Tests				10
4	Statistical Inference for Two samples	Inference on the Difference in Means of two Normal Distributions, A Nonparametric Test for the Difference in Two Means, Paired t-test, Inference on the variance				12

		of Two Normal Distributions, Inference on Two Population Proportions	
5	Simple Linear Regression	Simple Linear Regression, Least squares estimators, Hypothesis tests and Confidence Intervals in Single Linear Regression, Adequacy of the Regression Model: Residual Analysis, Coefficient of Determination, Regression on Transformed Variables, Logistic Regression, Polynomial Regression Models	15
		Total	56
Key Text : Montgomery Douglas C., Runger George C., Applied Statistics and Probability for Engineers, Seventh Edition, Wiley, 2018 Chapters: 7 -9 (up to 9.8), 10, 11 Suggested Readings: 1. Sheldon M Ross, Introduction to Probability and Statistics for Engineers and Scientists, Sixth Edition, Elsevier Academic Press, 2020 2. Irwin Miller, Marylees Miller, John E. Freund's Mathematical Statistics with Applications, Eighth Edition, Pearson Publications			

Course Title	Linear Algebra for Data Science	L	T	P	C	Periods
Course Code	UDSC-401	3	0	1	4	70
Course Category	IDM					
Course Objectives: This course provides an introduction to Linear algebra and its applications to other disciplines.						
Course Outcomes: By the end of the course, students will be able to: 1. Understand the use of linear algebra in modeling concepts. 2. Perform matrix calculations effectively. 3. Describe the concepts of vector spaces and their subspaces through various examples. 4. Differentiate between linearly independent and dependent sets and determine the basis for finite-dimensional vector spaces. 5. Analyze linear transformations between vector spaces, discern their properties using the image and kernel of the transformation. 6. Express linear transformations in matrix form and discuss their properties. 7. Understand the use of SVD. 8. Explain characteristic polynomials and comprehend the concepts of eigenvalues and eigenvectors in the context of endomorphisms.						
Prerequisites for the Course: 10+2 Mathematics, UDSC-101						
Unit	Topic	Contents				Periods.
1	Vector Spaces	Vectors - Linear Combinations, Lengths, Dot products. Matrices-Linear Equations, Elimination method, Rules of Matrix operations, Inverse, LU factorization, Transposes and permutations.				8
2	Fundamental Spaces	Vector space-Spaces of vectors, Null space, Complete solution to nonhomogeneous equation, Independence, Basis, Dimension. Subspaces - Dimension of Four subspaces, Orthogonality, Projection, Orthonormal bases, Gram Schmidt orthogonalisation				8
3	Determinants	Determinants-Properties, Permutations and Cofactors, Cramer's rule, Inverse, Volume. Eigenvalues and Eigenvectors - Eigenvalues, Diagonalisation, System of differential equations, Symmetric matrices, Positive Definite matrices				8
4	SVD	Singular Value Decomposition - Image processing, Bases and matrices, Principal Component Analysis, Geometry of SVD				10

5	Linear Transformations	Linear Transformations - Introduction, Matrix of Linear Transformation, Search for a good basis	8
		Total	42

Key Text

Gilbert Strang, Introduction to Linear Algebra, 5th Edition, Wellesley-Cambridge Press, 2016

Chapters: 1 (1.1-1.3), 2 (2.1-2.7), 3 (3.1-3.5), 4 (4.1-4.4), 5 (5.1-5.3), 6 (6.1-6.5), 7 (7.1-7.4)

https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/video_galleries/video-lectures/

Retrieved on Jan 31, 2023

REFERENCES

1. Stephen Boyd & Lieven Vandenberghe, Introduction to Applied Linear Algebra - Vectors, Matrices and Least Squares, Cambridge University Press, 2018
2. Philip N. Klein, Coding the Matrix - Linear Algebra through Applications to Computer Science, Newtonian Press, 2013

UDSC-401: Practicals: Linear Algebra for Data Science

Unit	Topic	Content	Periods
1	Python for Linear Algebra	The following programs are to be implemented in basic python - Dot Product of Vectors, Matrix Multiplication with basic python - Determinant of a Matrix, Matrix Inverse with basic python - Rank of a Matrix with basic python - Gaussian Elimination with basic python	12
2	Numpy and its usage in Linear Algebra	The following are to be implemented using numpy - Eigenvalues, Eigenvectors, SVD - Eigenfaces - Linear Regression - Matrix Exponential	16
		Total	28

Key Text: Leo Chin, Tanmay Dutta, NumPy Essentials, Packt Publishing, 2016

REFERENCES

Umit Mert Cakmak, Mert Cuhadaroglu, Mastering numerical computing with Numpy, Packt Publishing, 2018

Course Title	Optimization for Machine Learning	L	T	P	C	Periods
Course Code	UDSC-501	3	0	1	4	70
Course Category	IDM					
Course Objective: To modeling and discussion of documented real-world applications. Study of mathematical programming algorithms for optimization. Apply the mathematical results and numerical techniques of optimization theory to concrete Engineering problems.						
Course Outcome: Develop the skill set to 1. discovery, study and solve optimization problems; 2. Investigate, study, develop, organize and promote innovative solutions for various applications. 3. Apply mathematical skills to model optimization problems. 4. Critically analyze theoretical principles and choose the relevant optimization techniques for a specific problem. 5. Exposed to solutions for various types of optimization problems and will be enabled to adopt them for the situation at hand.						
Prerequisites for the Course: 10+2 Mathematics, Python programming, UDSC-101 Calculus						
Unit	Topic	Description				Periods
1	Introduction	Mathematical Review – Lines – Hyperplanes and linear varieties – convex sets – neighborhoods – polytopes and polyhedra – derivative matrix – level sets and gradients-Taylor’s series				6
2	Unconstrained Optimization	Unconstrained Optimization – conditions for local minimizers – one-dimensional search methods including golden section, Fibonacci, bisections and Newton’s method – Gradient methods including steepest descent methods, Newton's method and its analysis.				8
3	Linear Programing	Modeling with Linear Programming The Simplex method and Duality				8
4	Non-Linear Constrained Optimization	Non-linear constrained optimization – problems with equality constraints – Lagrange condition – second-order condition – minimizing quadratics subject to equality constraints – problems with inequality constraints – Karush-Kuhn-Tucker conditions – second-order conditions.				8

5	Convex Optimization	Convex Optimization Problems - Convex Functions - Convex Optimization Problems	6
6	Machine Learning view for Optimization	The minutiae of Gradient Descent, Properties of Optimization in ML, Computing Gradient wrt vectors, Linear Regression and Optimization models for Binary Targets	6
		Total	42

Suggested/Additional Readings:

1. L.R. Foulds, Optimization Techniques, Springer, Utm, 1981
2. <http://freevidelectures.com/Course/3072/Numerical-Optimization>
3. Hamdy A. Taha, Operations Research, Pearson Publication, 9th edition, (2014).
4. Boyd, Stephen, and Lieven Vanderberghe, Convex Optimization. Cambridge, UK: Cambridge University Press, 2004.

UDSC-501: Practicals - Optimization for Machine Learning

Unit	Topics	Contents	Periods
1	Unconstrained Optimization	Implementation of Unconstrained Optimization using Python: Newton-Raphson method; Implementation of Gradient Ascent and Gradient Descent methods	10
2	Solving LPP	Solving LP problems using Excel solver; Sensitivity Analysis using Excel solver; Solving LP problems using <i>scipy.optimize.linprog</i>	10
3	Python Library <i>scipy.optimize</i>	Univariate function minimization; Constrained and Unconstrained minimization of multivariate scalar function; Global Optimization; Least-Square Optimization; Custom minimizers	8
		Total	28

References:

1. SciPy Reference Guide, Retrieved July 7, 2022 from <https://docs.scipy.org/doc/scipy-1.7.1/scipy-ref-1.7.1.pdf>

Section 2.4 (Optimization - *scipy.optimize*)

2. Christian Hill, Learning Scientific Programming with Python, Second Edition, Cambridge University Press, 2020
3. Edwin K. P. Chong and Stanislaw H. Zak, An Introduction to Optimization, Fourth Edition, Wiley Inter - Science Publication, John Wiley and Sons Inc., 2013.
4. Hamdy A.Taha, Operations Research- An Introduction, 10th Edition, Pearson Education - 2017

Course Title	Data Mining and Machine Learning	L T P C Periods
Course Code	UDSC-502	3 0 1 4 70
Course Category	IDM	
Course Objectives: 1. To introduce students to the basic concepts and techniques of Machine Learning. 2. To become familiar with regression methods, classification methods, Pattern Mining and clustering methods. 3. To become familiar with Kernel Methods and Dimensionality reduction Techniques		
Course Outcomes: Develop the skill set to 1. Gain knowledge about basic concepts of Machine Learning 2. Identify machine learning techniques suitable for a given problem 3. Solve the problems using various machine learning techniques 4. Apply Dimensionality reduction techniques		
Prerequisites: UDSC-101/UMAT-101, UDSC-102, UDSC-201, UDSC-203 or Equivalent		
Unit	Title	Contents Periods
1	Introduction	Data Matrix, Numeric Attributes, Categorical Attributes, Graph Data, Kernel Methods, High-dimensional Data, Dimensionality Reduction, Types of Machine Learning8
2	Regression	Linear Regression, Logistic Regression, Neural Networks, Regression Evaluation8
3	Classification	Probabilistic Classification, Decision Tree Classifier, Linear Discriminant Analysis, Support Vector Machines, Classification Assessment.12
4	Clustering	Representative-based Clustering, Hierarchical Clustering , Density-based Clustering, Spectral and Graph Clustering, Clustering Validation14

5	Frequent pattern mining	Itemset Mining, Summarizing Itemsets Sequence Mining, Graph Pattern Mining.	14
Total			56
Key Text(s): 1. Mohammed J. Zaki, Wagner Meira, Jr., Data Mining and Machine Learning: Fundamental Concepts and Algorithms, 2nd Edition, Cambridge University Press, March 2020. 2. Trevor Hastie, Robert Tibshirani, and Jerome H. Friedman, The Elements of Statistical Learning, Second Edition, Springer, 2009 Retrieved on July 05, 2022 from https://hastie.su.domains/Papers/ESLII.pdf 4. Christopher M Bishop, Pattern Recognition and Machine Learning, Springer, 2006 5. Rogers and Girolami, A First Course in Machine Learning, Chapman and Hall/CRC, 2015			
References: 1. Barber, Bayesian Reasoning and Machine Learning, Cambridge University Press, 2012 2. Hal Daumé III, A Course in Machine Learning, e-Edition, 2012 3. Mitchell, Machine Learning, McGraw Hill, 1997 4. CS229-Machine Learning, Stanford University, Retrieved June 30, 2022 from https://see.stanford.edu/Course/CS229			
UDSC-502: Practicals- Data Mining and Machine Learning			
Unit	Title	Contents	Periods
1	Introduction	Review numpy, pandas and matplotlib for statistical, Probabilistic, algebraic and geometric views of data matrix Implementation of Linear algorithms, Non-Linear Algorithms and ensembling algorithms, bias-variance tradeoff, from scratch in python	12
2	Python Library scikit-learn	Develop different ML algorithms using scikit-learn	16
Key Text(s): 1. Jason Brownlee, Master Machine Learning Algorithms, Discover How They Work and Implement them from Scratch, Machine Learning Mastery, eBook, 2017 Retrieved June 30, 2022 https://github.com/AmandaZou/master-machine-learning-algorithms 2. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'reilly, 2019 3. Scikit-learn Documentation, retrieved June 30, 2022 from scikit-learn Tutorials — scikit-learn 1.0.2 documentation			

Course Title	Big Data Analytics	L	T	P	C	Periods
Course Code	UDSC - 601	4	0	0	4	56
Course Category	IDM					
Course Objectives: 1. To provide an overview of an exciting growing field of big data analytics. 2. To teach the fundamental techniques and principles in achieving big data analytics with scalability and streaming capability. 3. To enable students to have skills that will help them to solve complex real-world problems in decision support.						
Course Outcomes: Students will be able to 1. Understand the key issues in big data management and its associated applications in intelligent business and scientific computing. 2. Interpret business models and scientific computing paradigms, and apply software tools for big data analytics. 3. Achieve adequate perspectives of big data analytics in various applications like recommender systems, social media applications etc.						
Prerequisites for the Course: UDSC-302 Data Mining and Machine Learning						
S. No	UNIT	Topics				Periods
1	Introduction to Big Data	Big Data - Why and Where?, Characteristics of Big Data and Dimensions of Scalability, Getting value out of Big Data, Foundations for Big Data Systems and Programming				6
2	Similarity Algorithms	Near-Neighbor search, Shingling, Similarity preserving summary, Locality sensitive functions, Distance measures, Locality sensitive hashing and its applications to different distance measures, Applications of Locality sensitive hashing				12
3	Streaming Data	Stream Data model, Sampling data in a stream, Filtering streams, Counting distinct elements in a stream, Application of stream algorithms in counting.				10
4	Link Analysis	Page Rank, Computation of Page Rank, Topic sensitive page rank, Link spam.				6
5	Frequent Item sets	Market-Basket model, A-priori algorithm, Larger datasets in main memory, Limited pass algorithms, Counting frequent sets in a stream				10

6	Social Network Graphs	Clustering, Discovering of communities, Partitioning, Finding overlapping communities, Simrank, Counting Triangles, Neighborhood properties	12
		Total	56
Key Text(s): Anand Rajaraman, Mining Massive Datasets, Stanford University Press, 2014 Chapters: 2.1 - 2.3, 3.1 - 3.7, 4.1 - 4.6, 5.1 - 5.4, 6.1 - 6.4, 10			

Course Title	Cloud Computing	L	T	P	C	Periods
Course Code	UDSC-701	4	4	0	4	56
Course Category	IDM					

Course objectives:

This course gives students an insight into the basics of cloud computing along with virtualization, cloud computing is one of the fastest growing domains from a while now. It will provide the students basic understanding about cloud and virtualization along with how one can migrate over it.

Course outcomes: Develop the skill to

1. analyze the cloud computing setup with its vulnerabilities and applications using different architectures.
2. design different workflows according to requirements and apply map reduce programming model.
3. apply and design suitable virtualization concepts, cloud resource management and design scheduling algorithms.
4. create combinatorial auctions for cloud resources and design scheduling algorithms for computing clouds
5. assess cloud storage and security, the risks involved, its impact and develop cloud applications
6. broadly educate to know the impact of engineering on legal and social issues involved in addressing the security issues of cloud computing.

Prerequisites: Courses on Operating Systems, Computer Networks

Unit	Title	Contents	Periods
1	Cloud Computing Overview	Origins of Cloud computing – Cloud components - Essential characteristics – On-demand self service, Broad network access, Location independent resource pooling ,Rapid elasticity , Measured service, Comparing cloud providers with traditional IT service providers, Roots of cloud computing.	3
2	Cloud Insights	Architectural influences – High-performance computing, Utility and Enterprise grid computing, Cloud scenarios – Benefits: scalability ,simplicity ,vendors ,security, Limitations – Sensitive information - Application development- security level of third party - security benefits, Regularity issues: Government policies.	3

3	Cloud Architecture - layers and Models	Layers in cloud architecture, Software as a Service (SaaS), features of SaaS and benefits, Platform as a Service (PaaS), features of PaaS and benefits, Infrastructure as a Service (IaaS), features of IaaS and benefits, Service providers, challenges and risks in cloud adoption. Cloud deployment model: Public clouds	10
4	Cloud Simulators CloudSim and Green Cloud	Introduction to Simulator, understanding CloudSim simulator, CloudSim Architecture(User code, CloudSim, GridSim, SimJava) Understanding Working platform for CloudSim, Introduction to GreenCloud	20
5	Introduction to VMWare Simulator	Basics of VMWare, advantages of VMware virtualization, using VMware workstation, creating virtual machines-understanding virtual machines, creating a new virtual machine on local host, cloning virtual machines, virtualizing a physical machine, starting and stopping a virtual machine.	20
Total			56
Key Text: 1. Cloud computing a practical approach - Anthony T.Velte , Toby J. Velte Robert Elsenpeter, TATA McGraw- Hill , New Delhi – 2010 2. Cloud Computing: Web-Based Applications That Change the Way You Work and Collaborate Online - Michael Miller - Que 2008			
Reference Texts: 1. Cloud computing for dummies- Judith Hurwitz , Robin Bloor , Marcia Kaufman ,Fern Halper, Wiley Publishing, Inc, 2010 2. Cloud Computing (Principles and Paradigms), Edited by Rajkumar Buyya, James Broberg, Andrzej Goscinski, John Wiley & Sons, Inc. 2011			

Skill Enhancement Courses

PAPER CODE: UCSH-103		Software Lab in Python-I	CREDITS: 1 (0-0-2)
Course Category:		SEC	Total Periods: 28
Prerequisite (If any): NA			
Course Objectives: This course provides an introduction to the Python programming language. It covers the fundamentals of programming, including variables, data types, control structures, and functions. Students will learn how to write Python programs to solve simple problems and develop basic programming skills.			
Unit	Description		Periods
I	Introduction to Programming using Python Introduction to Programming Overview of the Python programming language Setting up the development environment Running Python program		2
II	Data Types and Operators int, float, complex, str, list, tuple, bytes, byte array, dictionaries Boolean type, Arithmetic operator, Relational operator, Logical or Boolean operator, Assignment Operator, Bitwise operator		8
III	Control Structures Conditionals: if, else, elif statements Loops: for and while loops Loop control statements: break, continue		8
IV	Functions Python built in functions, Python user defined functions, Python packages functions; Defining and calling Function, The anonymous Functions		10
	Total		28
Reference Text/Materials: 1. Mark Lutz, Learning Python, 5th Edition, O'Reilly Media Publishers 2. Think Python: How to think like a Computer Scientist by Allen Downey, 2nd edition, O'Reilly Publications: https://greenteapress.com/wp/think-python-2e/ https://www.w3schools.com/python			

PAPER CODE: UCSH-203	Software Lab in Python-II	CREDITS: 1((0-0-2)
Course Category:	SEC	Total Periods: 28
Prerequisite (If any): Software Lab in Python -I		
The course aims to: introduce students to advanced features of Python, such as complex data types, classes, and file handling, introduction to object-oriented Programming and data visualization		
Course Outcomes: At the end of the course, the student will be able to: • write and execute simple programs using strings, dictionaries, and tuples, and on classes and file handling in Python • create classes and objects • sketch graphs using appropriate Python libraries, and interpret them for a better understanding • develop proficiency in creating and manipulating NumPy arrays. • use mathematical and statistical operations on arrays.		
Unit	Description	Periods
I	Modules and Classes: How to write a module in Python, writing various customized modules for the purpose of reuse. Also introducing the concept of object-oriented programming using classes, inheritance, and encapsulation. Understanding List, Dictionary and Tuple Operations	10
II	Introduction to matplotlib (Using the pyplot submodule) figure, axes, subplot plot(), scatter(), show(), labels, legends, titles, styles, ticks, dynamically updating curves, saving graphs	6
III	Introduction to numpy The numpy array – properties: size, shape, ndim, dtype, creating arrays, zeros, one(), full(), fill(), arange(), linspace(), logspace(), identity(), eye(), astype(), indexing and slicing arrays (view versus copy), important array methods- reshape(), ravel(), flatten(), hstack() and vstack(), Aggregate functions- p.sum(), np.prod(), np.mean(), np.std(), np.var(), np.min(), np.max(), np.argmax(), np.argmin()	12
	Total	28
Reference Text/Materials: Reference Text/Materials: 1. Think Python: How to think like a Computer Scientist by Allen Downey, 2nd edition, O'Reilly Publications: https://greenteapress.com/wp/think-python-2e/ 2. Mark Lutz, Learning Python, 5th Edition, O'Reilly Media Publishers 3. https://www.w3resource.com/python/python-tutorial.php		

Course Title	Software Lab in Data Visualization	L	T	P	C	Periods
Course Code:	UCSH-303	0	0	4	2	56
Course Category	SEC					
Course Objectives: To make Data Storytelling through visualization of the data						
Course Outcomes: Develop the skill set to 1. Distinguish qualitative and quantitative data 2. Make inferences out of the different plots 3. Develop story out of the data						
Prerequisites for the Course: Basic Python/R						
Unit	Title Unit Contents					Periods
1	Numbers that Summarize the data	Measures of Average, Measures of Variance, Measures of correlation and Measures of Ratio.				14
2	Fundamental Variations of Graphs	A Brief History of Graphs; Graphical means for Qualitative data; Visual attributes: Lines, Histograms, Bar plot, Scatter plot and Box-Plots and inferences through examples				14
3	General Design Principles for Communication	Organizing, Highlighting, Integration, Table Design, General Graph Design, Multi-Variable display;				14
4	Case Studies	Take different data sets and make data storytelling with visualization.				14
		Total				56
*For implementation: Instructor can use any visualization framework Matplotlib/Seaborn/Plotly, ggplot2, Tableau, PowerBI, or any other						
Key Text: 1. Stephen Few, Show me the numbers: Designing tables and graphs to enlighten, Second Edition, Analytic Press, 2012 Chapters - 1 to 11 References: 2. Cole Nussbaumer Knafllic, Storytelling with Data, John Wiley & Sons, Inc. - 2015 3. Matplotlib Documentation, Retrieved June 30, 2022 from https://matplotlib.org/3.5.1/plot_CourseCategories/index.html 4. Seaborn Documentation, Retrieved June 30, 2022 from https://seaborn.pydata.org/introduction.ht						

Course Title	Software Lab in SQL		L	T	P	C	P
Course Code	UCSH-404		0	0	4	2	56
Course Category	SEC						
Course Objectives: It introduces basic query writing using SQL and MYSQL workbench. Course Outcome: Develop skill set to							
1. Write simple DDL statements using MySQL and MySQL Workbench 2. Write simple DML statements MySQL and MySQL Workbench 3. Implement triggers, logs along with locks							
Prerequisites for the Course: NA							
Unit	Topic	Contents	Periods				
1	SQL DDL (data Definition language)	Working on MYSQL command prompt as well as MYSQL workbench : To Create, insert ,delete, drop, alter, set, rename, operations on the databases (at least 6 tables from the databases mentioned) using with keys viz., primary, foreign and basic data Course Categories viz., varchar, int, float, date, time etc.	20				
2	SQL DML (Data manipulation languages)	MYSQL command prompt as well as MYSQL workbench : Using Select, from, where for single as well as multiples tables, Using Aliases, distinct,*,Union, minus and intersect erations, Nested queries, Exists function, use of NULL, Using aggregate functions viz, max, min and avg.	20				
3	Triggers	Triggers, Locks and Logs	16				
			Total			56	
References: 1. MYSQL reference manual 5.7. https://downloads.mysql.com/docs/workbench-en.pdf							

PAPER CODE: UMAT-003	FUNDAMENTALS OF ACTUARIAL MATHEMATICS	CREDITS: 3 (3-0-0)
Course Category:	SEC	Total Periods: 42
Pre-requisite (If any): NA		
Course Objectives: The aim of the Actuarial Mathematics subject is to provide a grounding in financial mathematics and its simple applications		
Course Outcomes: At the end of the course, the student will be able to: 1. Understand and be able to perform calculations relating to present value, current value, and accumulated value. 2. Calculate present value, current value, and accumulated value for sequences of non-contingent payments. 3. Understand key concepts concerning loans and how to perform related calculations. 4. Understand key concepts concerning bonds, and how to perform related calculations 5. Understand key concepts concerning yield curves, rates of return, measures of duration and convexity, cash flow matching and immunization, and how to perform related calculations.		
Unit	Description	Periods
I	Time Value of Money a. Define and recognize the definitions of the following terms: interest rate (rate of interest), simple interest, compound interest, accumulation function, future value, current value, present value, net present value, discount factor, discount rate (rate of discount), convertible monthly, nominal rate, effective rate, inflation and real rate of interest, force of interest, equation of value. b. Given any three of interest rate, period of time, present value, and future value, calculate the remaining item using simple or compound interest. Solve time value of money equations involving variable force of interest. c. Given any one of the effective interest rate, the nominal interest rate convertible m-thly, the effective discount rate, the nominal discount rate convertible m-thly, or the force of interest, calculate any of the other items. d. Write the equation of value given a set of cash flows and an interest rate.	7
II	Annuities/cash flows with non-contingent payments a. Define and recognize the definitions of the following terms: annuity-immediate, annuity due, perpetuity, payable m-thly or payable continuously, level payment annuity, arithmetic increasing/decreasing annuity, geometric increasing/decreasing annuity, term of annuity. b. For each of the following types of annuity/cash flows, given sufficient information of immediate or due, present value, future value, current value,	13

	interest rate, payment amount, and term of annuity, calculate any remaining item. - o Level annuity, finite term. - o Level perpetuity. - o Non-level annuities/cash flows. Arithmetic progression, finite term and perpetuity. Geometric progression, finite term and perpetuity. Other non-level annuities/cash flows.	
III	Loans a. Define and recognize the definitions of the following terms: principal, interest, term of loan, outstanding balance, final payment (drop payment, balloon payment), amortization. b. Calculate: • The missing item, given any four of: term of loan, interest rate, payment amount, payment period, principal. • The outstanding balance at any point in time. • The amount of interest and principal repayment in a given payment. • Similar calculations to the above when refinancing is involved	11
IV	Bonds a. Define and recognize the definitions of the following terms: price, book value, market value, amortization of premium, accumulation of discount, redemption value, par value/face value, yield rate, coupon, coupon rate, term of bond, callable/non-callable, call price, call premium, accumulated value with reinvestment of coupons. b. Given sufficient partial information about the items listed below, calculate any of the remaining items • Price, book value, market value, accumulated value with reinvestment of coupons, amortization of premium, accumulation of discount. (Note that valuation of bonds between coupon payment dates will not be covered). • Redemption value, face value, Yield rate, Coupon, coupon rate. • Term of bond, point in time that a bond has a given book value, amortization of premium, or accumulation of discount. Calculate the price of a callable bond to achieve a specified minimum yield.	11
Key Text: Broverman, S.A., Mathematics of Investment and Credit (Seventh Edition), 2017, ACTEX Publications, ISBN 978-1-63588-221-6		
Prescribed Chapters/ Sections : Chapter 1 (excluding 1.2.1 and 1.8), Chapter 2 (excluding 2.3.1.2, 2.4.2, 2.4.3 and 2.4.5), Chapter 3 (excluding 3.2.1, 3.2.2, 3.3, and 3.4), Chapter 4 (excluding 4.1.3), Chapter 5 (excluding 5.2, the investment year method portion of 5.3.1, and excluding all of 5.3.2, 5.3.3 and 5.3.4), Chapter 6 (excluding 6.2 and 6.4), Chapter 7 (excluding 7.1.3, 7.1.6 and 7.3)		
Suggested Readings: 1. Bowers, N. L.; Gerber, H. U.; Hickman, J, C. et al., In Actuarial Mathematics, 2nd ed. Society of Actuaries, 1997. 753 pages. ISBN: 9780938959465. 2. Butcher, M. V., Mathematics of compound interest, Nesbitt, C. J. Ulrich's Books, 1971. 324 pages. ISBN: 9780960300013.		

3. Ingersoll, J. E. Rowman & Littlefield, 1987, Theory of financial decision making. 474 pages. ISBN: 9780847673599.
4. Kellison, S. G., The theory of interest. 3rd ed. Irwin, 2008. 463 pages. ISBN: 9780073382449.
5. Gerber, H. U., Life insurance mathematics, 3rd ed. Springer; Swiss Association of Actuaries, 1997. 217 pages. ISBN: 9783540622420.

Multi-Disciplinary Courses

PAPER CODE: UMAT-001	Fundamentals of Statistics with Excel	CREDITS: 3 (2-0-1)
Course Category:	MDC	Total Periods: 42
Pre-requisite (If any): 10+2 any Discipline		
Course Objectives: The objective is to give basic knowledge of the subject and familiarity with techniques to use them effectively. Knowledge of Mathematics at the Senior Secondary level will be an advantage but not necessary for offering this course. The scope of Statistics has been well recognized over the years but has very much expanded in recent years and with the impressive progress in		
Course Outcomes: At the end of the course, the student will be able to: 1. Recognize and differentiate between key terms. 2. Apply various types of sampling methods to data collection. 3. Create and interpret frequency tables. 4. Display data graphically and interpret graphs: stemplots, histograms, and box plots. 5. Recognize, describe, and calculate the measures of location of data: quartiles and percentiles. 6. Recognize, describe, and calculate the measures of the center of data: mean, median, and mode. 7. Recognize, describe, and calculate the measures of the spread of data: variance, standard deviation, and range.		
Unit	Description	Periods
1	Statistics: What is statistics, Uses and misuses of statistics, Introduction to basic terms, Measurability and variability, Relationship between probability and statistics?	4
2	Descriptive Analysis and Presentation of Single Variable Data: Data collection, Measures of central tendency, Measures of Dispersion, Graphic Presentation and Measures of Position.	14
3	Descriptive Analysis and Presentation of Single Variable Data (Continued)	12

	Frequency distributions, Graphic presentations, Calculations using frequency distribution, Interpreting and understanding Standard Deviation, The art of Statistical Deception.	
4	Descriptive Analysis and Presentation of Bivariate Data Bivariate Data, Linear Correlation, Linear Regression	12
Key Text: 1. Robert R. Johnson, Elementary Statistics, Duxbury Press, North Scituate, Massachusetts, 1976, 2 nd edition Chapters: 1,2,3, 4		
Suggested Readings: 1. Mario F. Triola, Elementary Statistics, Addison-Wesley, 7 th Edition, 1997. 2. Robert N. Goldman, Joel S. Weinberg, Statistics An Introduction, Prentice-Hall Publication, 1985.		

PAPER CODE: UMAT-002	The Language of Mathematics	CREDITS: 3 (3-0-0)
Course Category:	MDC	Total Periods: 42
Prerequisite (If any): 10+2 Mathematics		
Course Objectives: The course aims to: 1. Give a basic understanding of working with numbers which include Modular Arithmetic, the Principle of Induction, and the usage of prime numbers. 2. Give a basic understanding of Geometry, Euclid's elements, conic sections, non-Euclidean geometry, and higher dimensions 3. Give a basic understanding of Probability, experiments, and chance		
Course Outcomes: At the end of the course, the student will be able to: 1. Start any abstract course on Number Theory 2. Read courses on Applied and Theoretical Probability.		
Unit	Description	Periods
I	Why Do Numbers Count? You can count on them; These days, children do it before they are five; A token advance; Symbolic Progress; For a long time, it was all Greek; A fatal law is discovered; Here's looking at Euclid; Numbers in Prime condition; Prime order; The child Genius; Gauss's clock Arithmetic; The Great Amateur; Taking the prime test; Keeping Secrets; Easy to Guess, Hard to prove; Fermat's Last Theorem; The Fermat Saga Begins; The Domino Effect	18
II	Mathematics Gets into Shape Everyone is a geometer; Measuring the Earth; Things that Euclid missed; Euclid in his elements; Geometry as a theory of Everything; Slicing through Cones; A fly on the ceiling; Squaring the circle and other impossibilities; The startling discovery of non-Euclidean geometries; The geometry of Renaissance artists; Beyond the third dimension	18
III	How do mathematicians figure the odds? Who gets Heaven?; Figuring the odds; The geometric patterns of chance; In Mathematics we trust; The Amazing Bernoulli; Fear of Flying; Ringing in the bell curve; The Reverend Bayes; Enter the Average Man; The abstract Patterns of chance; Using Mathematics to choose your options	20
Key Text:		
1. Keith Devlin, The Language of Mathematics (Making the invisible visible), W H Freeman and Company, 2000, 1 st edition		Numbers, Geometry and Probability
Suggested Readings:		Chapter: 1, 4, 7
1. Richard Courant and Herbert Robbins, what is Mathematics? Oxford University Press, 1941, 2 nd Edition		